

TRACK AND TRACE SUPPLYCHAIN

A PROJECT REPORT

Submitted by,

K V BRAMHANANDA REDDY	- 20211CSE0148
THOTA YASWANTH RAM	- 20211CSE0093
KUMMARA CHARAN SAI	- 20211CSE0097
D GURUVARDHAN REDDY	- 20211CSE0121

Under the guidance of,

Ms. AYESHA TARANUM

in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

At



PRESIDENCY UNIVERSITY

BENGALURU

DECEMBER 2024

PRESIDENCY UNIVERSITY

SCHOOL OF COMPUTER SCIENCE ENGINEERING

CERTIFICATE

This is to certify that the Project report “**TRACK AND TRACE SUPPLYCHAIN**” being submitted by “K V BRAMHANANDA REDDY” bearing roll number(s) “20211CSE0148”, by “THOTA YASWANTH RAM” bearing roll number(s) “20211CSE0093”, by “KUMMARA CHARAN SAI” bearing roll number(s) “20211CSE0097”, by “D GURUVARDHAN REDDY” bearing roll number(s) “20211CSE0121” in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in **Computer Science and Engineering** is a bonafide work carried out under my supervision.

Ms. Ayesha Taranum
Assistant Professor
School of CSE&IS
Presidency University

Dr. Asif Mohammed
Associate Professor & HoD
School of CSE&IS
Presidency University

Dr. L. SHAKKEERA
Associate Dean
School of CSE
Presidency University

Dr. MYDHILI NAIR
Associate Dean
School of CSE
Presidency University

Dr. SAMEERUDDIN KHAN
Pro-Vc School of Engineering
Dean -School of CSE&IS
Presidency University

PRESIDENCY UNIVERSITY

SCHOOL OF COMPUTER SCIENCE ENGINEERING

DECLARATION

We hereby declare that the work, which is being presented in the project report entitled **TRACK AND TRACE SUPPLYCHAIN** in partial fulfillment for the award of Degree of **Bachelor of Technology** in **Computer Science and Engineering**, is a record of our own investigations carried under the guidance of **Ms. Ayesha Taranum, Assistant Professor, School of Computer Science Engineering & Information Science, Presidency University, Bengaluru.**

We have not submitted the matter presented in this report anywhere for the award of any other Degree.

Names	Roll Numbers	Signatures
K V Bramhanada Reddy	20211CSE0148	
Thota Yaswanth Ram	20211CSE0093	
Kummara Charan Sai	20211CSE0097	
D Guruvardhan Reddy	20211CSE0121	

ABSTRACT

In today's interconnected global economy, the complexity of supply chains has grown exponentially, involving multiple stakeholders, diverse geographies, and intricate processes. These complexities, coupled with the increasing demand for transparency and efficiency, have highlighted the need for a robust Track and Trace Supply Chain System. This project is designed to address these challenges by integrating advanced technologies such as the Internet of Things (IoT), Radio Frequency Identification (RFID), GPS tracking, blockchain, and artificial intelligence (AI). Together, these tools enable seamless tracking and tracing of goods from their point of origin to their final destination, ensuring transparency, accountability, and operational efficiency throughout the supply chain.

Track and trace systems have emerged as an indispensable component of modern supply chain management, addressing the growing demand for transparency, efficiency, and security. These systems enable businesses to monitor and document the movement of goods at every stage of their journey, from raw material sourcing to final delivery. By utilizing technologies such as barcodes, QR codes, RFID tags, IoT sensors, GPS, and blockchain, track and trace systems offer real-time visibility and ensure accountability. They are particularly valuable in industries where quality, compliance, and safety are critical, such as pharmaceuticals, food, automotive, and retail.

The system leverages IoT and RFID technology to continuously monitor and record critical parameters such as temperature, humidity, and location. This ensures that goods are maintained under optimal conditions, especially in sensitive industries like pharmaceuticals and food. GPS tracking combined with geo-fencing provides real-time location updates, alerts for unauthorized deviations, and route optimization. Blockchain technology plays a vital role in maintaining data integrity by creating immutable and tamper-proof records, fostering trust among stakeholders. Smart contracts further automate critical processes such as payments and compliance checks, reducing the likelihood of disputes and delays.

Global supply chains have become increasingly complex due to factors like globalization, outsourcing, and diverse regulatory requirements. This complexity creates risks such as counterfeiting, theft, inefficiencies, and product recalls. Track and trace systems address these challenges by enabling businesses to track the current location of products and trace their history in case of issues. They also support compliance with stringent regulations, improve customer trust, and optimize logistics.

AI and machine learning are at the core of the system's predictive capabilities. By analyzing historical data and current trends, these technologies provide insights into potential risks and recommend proactive measures to mitigate them.

ACKNOWLEDGEMENT

First of all, we indebted to the **GOD ALMIGHTY** for giving me an opportunity to excel in our efforts to complete this project on time. We express our sincere thanks to our respected dean **Dr. Md. Sameeruddin Khan**, Pro-VC, School of Engineering and Dean, School of Computer Science Engineering & Information Science, Presidency University for getting us permission to undergo the project. We express our heartfelt gratitude to our beloved Associate Deans **Dr. Shakkeera L** and **Dr. Mydhili Nair**, School of Computer Science Engineering & Information Science, Presidency University, and Dr .Asif Mohammed Head of the Department, School of Computer Science Engineering & Information Science, Presidency University, for rendering timely help in completing this project successfully. We are greatly indebted to our guide **Ms.Ayesha Taranum** Assistant Professor and Reviewer **Dr.Mr.Jayachandran**, School of Computer Science Engineering & Information Science, Presidency University for his inspirational guidance, and valuable suggestions and for providing us a chance to express our technical capabilities in every respect for the completion of the project work. We would like to convey our gratitude and heartfelt thanks to the PIP2001 Capstone Project Coordinators **Dr. Sampath A K**, **Dr. Abdul Khadar A** and **Mr. Md Zia Ur Rahman**, department Project Coordinators **Mr.Amarnath J L** and Git hub coordinator **Mr.Muthuraj**. We thank our family and friends for the strong support and inspiration they have provided us in bringing out this project.

K V Bramhanada Reddy

Thota Yaswanth Ram

Kummara Charan Sai

D Guruvardhan Reddy

LIST OF TABLES

Sl. No.	Table Name	Table Caption	Page No.
1	9.3	Test Cases	39

LIST OF FIGURES

Sl. No.	Figure Name	Caption	Page No.
1	Figure 3.1.3	Project Flow	17
2	Figure 6.3.1	Use case Diagram	26
3	Figure 6.3.2	Class Diagram	26
4	Figure 6.3.3	Sequence Diagram	27
5	Figure 6.3.4	Collaboration Diagram	28
6	Figure 6.3.5	Activity Diagram	29
7	Figure 6.3.6	Deployment Diagram	30
8	Figure 6.3.7	Component Diagram	30
9	Figure 6.4	Data Flow Diagram	31
10	Figure 6.5	Architecture Diagram	32

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	iv
	ACKNOWLEDGEMENT	v
	LIST OF TABLES	vi
	LIST OF FIGURES	vii
1	INTRODUCTION	
	1.1 Overview	1
	1.2 The Importance of Track and Trace in Supply Chains	
	1.2.1 Enhancing Visibility and Transparency	3
	1.2.2 Mitigating Risks and Ensuring Product Safety	3
	1.2.3 Streamlining Recall Management	4
	1.2.4 Facilitating Compliance with Regulations	5
	1.2.5 Improving Operational Efficiency	6
	1.3 Key Technologies Enabling Track and Trace	
	1.3.1 Radio Frequency Identification (RFID)	7
	1.3.2 Barcode Technology	8
	1.3.3 Blockchain Technology	8
	1.4 Future Trends in Track and Trace	
	1.4.1 Artificial Intelligence (AI) and Machine Learning	9
	1.4.2 Internet of Things (IoT)	9
	1.5 Benefits of Track and Trace Supply Chain Systems	
	1.5.1 Improved Visibility and Transparency	10
	1.5.2 Improved Efficiency and Productivity	10
2	LITERATURE SURVEY	
	2.1 Related Work	11
3	RESEARCH GAPS OF EXISTING METHODS	

4	3.1 SYSTEM ANALYSIS	
	3.1.1 Existing System	16
	3.1.2 Proposed System	16
	3.1.3 Project flow:	17
	3.2 REQUIREMENT ANALYSIS:	
	3.2.1 Function and non-functional requirements	17
	3.2.2 Hardware Requirements	18
	3.2..3 Software Requirements	18
	PROPOSED METHODOLOGY	
	4.1 Smart Contracts	19
5	4.2 Keccak Hashing	19
	4.3 Identification and Data Capture	20
	4.4 Sustainability Tracking	20
	4.5 Enhanced Customer Experience	20
	4.6 Real-Time Notifications and Alerts	21
5	4.7 Product authentication	21
	OBJECTIVES	
	5.1 Achieve End-to-End Supply Chain Visibility	22
	5.2 Integrate Cost-Effective Solutions for SMEs	22
	5.3 Automate Processes with Smart Contracts	22
6	5.4 Enhance Risk Management and Anomaly Detection	23
	5.5 Facilitate Regulatory Compliance	23
	SYSTEM DESIGN & IMPLEMENTATION	
	6.1 Introduction of Input design	24
	6.2 Output Design	24
	6.3 UML diagrams	
	6.3.1 USE CASE DIAGRAM	25
	6.3.2 CLASS DIAGRAM	26
	6.3.3 SEQUENCE DIAGRAM	27
	6.3.4 COLLABORATION DIAGRAM	28
	6.3.5 ACTIVITY DIAGRAM	29
	6.3.6 DEPLOYMENT DIAGRAM	30

	6.3.7 COMPONENT DIAGRAM	30
	6.4 Data Flow diagrams	31
	6.5 Architecture Diagram	32
7	TIMELINE FOR EXECUTION OF	33
8	PROJECT OUTCOMES	34
9	RESULTS AND DISCUSSIONS	
	9.1 System Study	
	9.1.1 Economic Feasibility	37
	9.1.2 Technical Feasibility	37
	9.1.3 Social Feasibility	38
	9.2 System Testing	
	9.2.1 Unit Testing	38
	9.2.2 Integration Testing	38
	9.2.3 Functional Testing	38
	9.2.4 White Box Testing	39
	9.2.5 Black Box Testing	39
	9.3 Test cases	39
10	CONCLUSION	40
	REFERENCES	41
	APPENDIX-A	42
	APPENDIX-B	44
	APPENDIX-C	48

CHAPTER-1

INTRODUCTION

1.1 Overview

Track and Trace is a system that enables businesses to monitor the movement of goods throughout their supply chain journey, from origin to destination. By utilizing technologies like barcodes, RFID, GPS, and IoT sensors, businesses can capture and analyze data at every stage, providing real-time visibility and control over their supply chain. This data-driven approach offers numerous benefits, including improved efficiency, reduced costs, enhanced customer satisfaction, and enhanced compliance. However, implementing track and trace systems can be complex and costly, requiring significant investment in technology, infrastructure, and skilled personnel. Additionally, challenges such as data security, interoperability, and resistance to change can hinder adoption. To overcome these obstacles and maximize the potential of track and trace, businesses need to embrace emerging technologies like blockchain, AI, and machine learning, as well as foster a culture of innovation and collaboration across the entire supply chain.

Track and trace refers to the process of monitoring and recording the movement of goods, products, or assets through various stages of the supply chain. This involves tracking the origin, movement, storage, and delivery of products, as well as identifying any discrepancies or irregularities. The primary goal of track and trace is to ensure the efficient and secure movement of goods, while also enabling real-time monitoring and response to any potential issues.

In the context of evolving global supply chains, the importance of track and trace systems continues to grow. With increasing consumer demand for transparency and sustainability, businesses are challenged to provide proof of responsible sourcing and ethical practices. Companies are now not only accountable to regulators but also to a savvy consumer base that values transparency regarding where and how products are made. This is where advanced technologies, including blockchain, come into play, offering decentralized, tamper-proof records of the supply chain. By integrating these innovations, companies can not only streamline their operations but also meet rising consumer expectations for sustainability, ethical sourcing, and social responsibility.

Moreover, the growing complexity of supply chains, exacerbated by global events

such as pandemics or geopolitical tensions, makes the need for robust track and trace systems even more pronounced. Companies are recognizing that having a comprehensive tracking system can aid in risk management, enabling them to anticipate disruptions and develop contingency plans. For instance, if a specific ingredient in food products is linked to contamination, a strong track and trace system can quickly identify the source and all affected products, facilitating swift recalls and minimizing consumer risk.

As we move further into a digital era, the integration of artificial intelligence (AI) and machine learning (ML) with track and trace systems is poised to revolutionize supply chain management. These technologies can analyse vast amounts of data to identify patterns, predict potential issues, and optimize routes, contributing to more efficient operations. Additionally, predictive analytics can empower companies to make more informed decisions about inventory levels and sourcing strategies, thus driving profitability.

In conclusion, track and trace systems represent a vital component of modern supply chain management. They not only ensure compliance and operational efficiency but also enhance customer trust and brand loyalty. As technology continues to evolve, businesses that prioritize the development and implementation of sophisticated track and trace solutions will be better positioned to navigate the complexities of global supply chains and meet the demands of today's consumers. Embracing these advancements will ultimately lead to a more resilient, transparent, and sustainable supply chain, aligning with the broader goals of economic and environmental stewardship in the contemporary market landscape.

The benefits of implementing effective track and trace systems in the supply chain are extensive. Firstly, they ensure better inventory management by allowing for precise tracking of stock levels, reducing the risk of overstocking or stockouts. Real-time tracking helps organizations optimize their logistics and distribution strategies, resulting in reduced operational costs. Furthermore, this transparency aids in the quick identification of issues—such as delays, losses, or defects—enabling companies to respond promptly. In industries like pharmaceuticals and food, where safety and regulatory compliance are imperative, track and trace technologies help ensure product authenticity and adherence to stringent regulations, thus protecting consumers and maintaining brand integrity. Additionally, track and trace can enhance customer satisfaction by providing consumers with information regarding the origin and journey of their products, fostering trust in the brand.

1.2 The Importance of Track and Trace in Supply Chains

1.2.1 Enhancing Visibility and Transparency

Track and trace systems play a critical role in providing end-to-end visibility across the supply chain. In today's globalized economy, where products pass through multiple touchpoints, stakeholders need real-time access to information about the movement and status of goods. This transparency helps businesses identify bottlenecks, inefficiencies, and potential risks early, ensuring smoother operations. For instance, a manufacturer can track raw materials to ensure timely production schedules, while retailers can monitor inventory levels to prevent stockouts or overstocking. Such visibility fosters better collaboration between suppliers, manufacturers, distributors, and customers, driving overall efficiency.

Enhancing visibility and transparency in track and trace supply chains is crucial for businesses to thrive in today's competitive market. By implementing robust track and trace systems, fostering collaboration and information sharing, leveraging data analytics, and embracing digital transformation, businesses can gain a competitive edge. Real-time monitoring of shipments, improved decision-making, enhanced customer satisfaction, and strengthened brand reputation are just a few of the benefits that can be achieved through increased visibility and transparency. Additionally, by adhering to regulatory requirements and industry standards, businesses can ensure compliance and mitigate risks.

1.2.2 Mitigating Risks and Ensuring Product Safety

In industries such as pharmaceuticals, food, and electronics, the prevalence of counterfeit goods has emerged as a serious challenge, jeopardizing consumer safety, trust, and brand integrity. The repercussions of counterfeit products are particularly alarming in pharmaceuticals, where fake medications can lead to ineffective treatment or even fatal health consequences. Similarly, counterfeit food products can pose health risks through contamination or improper labeling, while counterfeit electronic goods can malfunction or pose safety hazards. To combat these threats, track and trace systems have become invaluable. These systems utilize unique identifiers like barcodes and RFID (Radio Frequency Identification) tags, which are assigned to products at various stages of production and distribution. By integrating these identifiers into the supply chain, businesses can monitor a product's journey from its point of origin to the consumer, thereby confirming its authenticity.

The implementation of track and trace solutions not only mitigates the circulation of counterfeit and substandard goods but also enhances transparency and accountability within the supply chain. For example, if a counterfeit product is detected, businesses can quickly

trace its source, allowing them to quarantine affected batches and alert customers. Furthermore, these systems support regulatory compliance by providing necessary documentation and trail records to authorities, which is essential in industries where safety standards are strictly enforced.

Additionally, track and trace systems can significantly improve temperature and environmental monitoring for sensitive items, such as pharmaceuticals and certain food products that require specific storage conditions. Advanced sensors and IoT (Internet of Things) technology can be integrated with these systems to monitor temperature, humidity, and other environmental factors in real time. This proactive approach helps ensure that perishable goods are kept within their optimal ranges, safeguarding quality and preventing spoilage or degradation. In essence, track and trace systems are pivotal in not only combating counterfeiting but also enhancing the overall safety, quality, and reliability of products across critical industries, thereby fostering consumer confidence and protecting public health.

1.2.3 Streamlining Recall Management

Streamlining recall management within the track and trace supply chain is crucial for enhancing safety, compliance, and operational efficiency. The ability to swiftly identify, trace, and manage product recalls is increasingly essential for businesses across various sectors, particularly in food, pharmaceuticals, and consumer goods. With the advent of rigorous regulatory standards and heightened consumer awareness, companies must optimize their recall processes to minimize risks associated with defective or harmful products.

A robust track and trace system serves as the backbone of effective recall management. By leveraging advanced technologies such as RFID (radio-frequency identification), barcoding, and blockchain, businesses can obtain real-time visibility into their inventory. This visibility enables quick identification of affected products within the supply chain, allowing for rapid response actions when a recall is initiated. Implementing a seamless track and trace system enhances the accuracy of data collection on product movements, thereby reducing the time it takes to notify stakeholders and recover defective items.

Moreover, the integration of digital platforms can improve communication among manufacturers, distributors, and retailers during a recall. Automated notifications and reporting systems can keep all parties informed and facilitate coordinated efforts to manage the recall process. This interconnectedness not only accelerates the retrieval of products from the market but also reassures consumers through timely updates, fostering trust and transparency. The collaborative approach allows companies to act decisively and efficiently, minimizing the potential fallout from a recall situation.

Additionally, data analytics plays a significant role in streamlining recall management. By analyzing historical data on product performance, consumer feedback, and incident reports, companies can identify recurring issues or trends that may lead to future recalls. Predictive analytics can aid in forecasting potential supply chain vulnerabilities, allowing organizations to implement preventive measures proactively. This proactive stance not only mitigates risks but also enhances overall product quality and safety in the long run.

1.2.4 Facilitating Compliance with Regulations

Facilitating compliance with regulations in the context of a track and trace supply chain is a multifaceted endeavour that requires a thorough understanding of both the regulatory landscape and the capabilities of modern tracking technologies. In industries such as food, pharmaceuticals, and consumer goods, meeting stringent regulatory requirements is crucial not only for legal compliance but also for ensuring product safety and consumer trust. Implementing an effective track and trace system serves as a cornerstone for achieving and maintaining compliance across the supply chain.

At the heart of compliance is the ability to provide verifiable records of a product's journey from origin to end consumer. Regulations like the Food Safety Modernization Act (FSMA) in the United States and the European Union's General Food Law emphasize the importance of traceability in food supply chains. To achieve compliance, businesses must establish a comprehensive documentation process that logs every step of the supply chain, including sourcing, processing, and distribution. Integrating technologies such as barcodes, RFID tags, and IoT devices can facilitate real-time data capture, ensuring that the necessary information is readily available for audits and inspections.

Furthermore, data integrity is paramount for regulatory compliance. Organizations must not only collect data but also ensure its accuracy and consistency throughout the supply chain. Utilizing blockchain technology can provide a secure and tamper-proof method for recording transactions and product movements. Each change in product status or location is logged in a decentralized ledger, creating a transparent trail that regulators can access for verification purposes. This level of transparency not only strengthens compliance but also enhances operational efficiency by minimizing disputes and errors related to product provenance.

In addition to tracking and documenting product movements, compliance with regulations often requires organizations to have robust recall mechanisms in place. Effective track and trace systems enable companies to rapidly identify and locate affected products in the event of a safety issue. By streamlining recall processes, businesses can not only fulfil

regulatory obligations to notify authorities and consumers promptly but also mitigate risks associated with public health and safety incidents. Having predetermined protocols for recalls enhances readiness and compliance while minimizing the potential for reputational damage.

Training and awareness are also critical factors in facilitating compliance within the supply chain. Employees at all levels must be educated about relevant regulations and the importance of adherence to standardized processes. Regular training sessions can help reinforce the significance of traceability and compliance, ensuring that staff understands their role in the overarching supply chain system. By creating a culture of compliance, organizations can empower their workforce to prioritize regulatory requirements and contribute to the overall integrity of the supply chain.

1.2.5 Improving Operational Efficiency

Improving operational efficiency in the track and trace supply chain is essential for businesses striving to meet customer demands while optimizing costs and resource utilization. The integration of advanced technologies and streamlined processes within the supply chain can significantly enhance visibility, accuracy, and responsiveness, ultimately leading to greater operational efficiency. This approach not only helps in minimizing delays and reducing waste but also positions companies competitively in their respective markets.

One of the fundamental ways to improve operational efficiency is through real-time visibility across the supply chain. Implementing RFID technology, barcoding, and IoT-enabled devices allows organizations to monitor inventory levels, track product movements, and gather data on supply chain performance instantaneously. The ability to access real-time information empowers decision-makers to respond quickly to changes or disruptions in the supply chain, such as unexpected demand fluctuations or supply shortages. By reducing lead times and improving the accuracy of inventory management, businesses can make more informed decisions that optimize resources and increase overall productivity.

Additionally, automated data capture and processing play a vital role in enhancing operational efficiency. Manual data entry can be time-consuming and prone to errors, leading to inaccuracies that can disrupt the supply chain. Implementing automated systems for capturing and processing data minimizes human intervention and reduces the risk of mistakes. This automation not only speeds up processes but also ensures that stakeholders have access to accurate and up-to-date information. For instance, automated inventory tracking can help organizations maintain optimal stock levels, preventing overstocking or stockouts, which ultimately contributes to cost savings.

Furthermore, leveraging data analytics within the track and trace framework can

provide valuable insights into supply chain performance and areas for improvement. By analysing historical data patterns, organizations can identify bottlenecks, inefficiencies, and trends that may impact operations. Predictive analytics can help forecast demand more accurately, optimize inventory levels, and streamline logistics planning. These insights allow organizations to proactively address issues, enhance delivery efficiency, and reduce operational costs, resulting in a more efficient supply chain overall.

Collaboration and communication among supply chain partners are also key elements in improving operational efficiency. An effective track and trace system facilitates seamless information sharing between manufacturers, suppliers, distributors, and retailers. Enhanced communication can lead to better coordination of activities, aligning production schedules, and reducing the risk of miscommunication that could lead to delays. When all stakeholders have access to the same real-time data, it fosters a collaborative environment where decisions can be made based on accurate information, ultimately leading to smoother workflows and improved operational performance.

1.3 Key Technologies Enabling Track and Trace

1.3.1 Radio Frequency Identification (RFID)

Radio Frequency Identification (RFID) technology has emerged as a transformative tool in enhancing track and trace capabilities within supply chains. By utilizing electromagnetic fields to automatically identify and track tags attached to objects, RFID offers significant advantages over traditional tracking methods such as barcodes. This technology enables real-time visibility into the movement and status of inventory, which is crucial for effective supply chain management.

One of the primary benefits of RFID in the track and trace supply chain is its ability to provide real-time data on inventory levels and product locations. Unlike barcodes, which require line-of-sight scanning, RFID tags can be read from a distance and do not require manual intervention to be scanned. This capability allows for continuous tracking of products as they move through various stages of the supply chain, from manufacturing and warehousing to distribution and retail. The result is improved accuracy in inventory counts, reduced discrepancies, and enhanced operational efficiency.

Additionally, RFID technology facilitates better inventory management by enabling automatic stocktaking and replenishment processes. With RFID systems, businesses can conduct real-time inventory audits without needing to halt operations for manual scanning.

1.3.2 Barcode Technology

Barcode technology has been a foundational element in supply chain management for decades, providing a simple yet powerful means of tracking and tracing products through various stages of the supply chain. Leveraging a two-dimensional matrix or a one-dimensional line code, barcodes encode information that can be quickly scanned and interpreted by barcode readers. This technology plays a critical role in enhancing operational efficiency, accuracy, and transparency across supply chain processes.

One of the primary advantages of barcode technology is its ability to facilitate quick and efficient data capture. By attaching barcodes to products, shipments, and inventory items, businesses can streamline the process of recording and managing data. Scanning a barcode allows for rapid identification and entry of product information into inventory management systems, reducing the time spent on manual data entry. The integration of barcode technology with other systems enhances its value in track and trace applications. For instance, when combined with warehouse management systems (WMS) and enterprise resource planning (ERP) solutions, barcode scanning can facilitate seamless data flow between various operational areas.

1.3.3 Blockchain Technology

Blockchain technology has the potential to revolutionize track and trace capabilities within supply chains, offering a secure and transparent way to document the journey of products from origin to end consumer. Unlike traditional databases that are managed by a single entity, blockchain operates on a decentralized network where data is recorded in a distributed ledger format. This approach enhances the integrity, visibility, and accountability of supply chain processes, ultimately improving operational efficiency and consumer trust.

Despite its numerous advantages, the implementation of blockchain technology in supply chains is not without challenges. One significant hurdle is the need for industry-wide collaboration and standardization. For blockchain to be effective, all participants in the supply chain must adopt compatible systems and practices. Additionally, the initial investment in developing and integrating blockchain solutions can be substantial, especially for small and medium-sized enterprises. Blockchain technology can streamline supply chain processes by reducing the complexity and time associated with traditional documentation methods. Traditionally, tracking products often involves numerous paper documents and multiple parties exchanging information, leading to delays and potential miscommunication.

1.4 Future Trends in Track and Trace

1.4.1 Artificial Intelligence (AI) and Machine Learning

One of the primary benefits of AI and ML in the track and trace supply chain is their ability to analyze large datasets in real-time. In a typical supply chain, numerous data points are generated at various stages, including production, warehousing, transportation, and sales. AI algorithms can process this data rapidly, identifying patterns and anomalies that might go unnoticed with traditional analytical methods. For instance, AI can correlate shipment delays with traffic patterns, weather conditions, or even historical performance, allowing businesses to proactively address potential disruptions.

AI and ML also significantly enhance the capabilities of supply chain visibility tools. Traditional methods for tracking products often involve manual data entry and sometimes result in gaps in information. However, AI can integrate data from various sources—such as RFID, IoT sensors, and ERP systems providing a comprehensive view of product movements and conditions throughout the supply chain.

1.4.2 Internet of Things (IoT)

The Internet of Things (IoT) is revolutionizing the track and trace capabilities of supply chains by connecting physical objects to the internet, enabling real-time data collection and communication. IoT devices, such as sensors, RFID tags, and smart containers, facilitate the continuous monitoring of products as they move through the supply chain. This technology enhances visibility, improves operational efficiency, and enables more informed decision-making, fundamentally transforming how businesses manage their supply chains.

Another crucial application of IoT in track and trace supply chain management is the automation of data collection processes. Traditionally, tracking products through manual entry and paperwork was prone to errors and inefficiencies. IoT devices automate this process by continuously collecting and transmitting data, which can then be integrated into supply chain management systems. IoT technology also supports predictive analytics within the supply chain. By collecting data on various parameters, such as transport times, environmental conditions, and consumer behavior, IoT devices can feed this information into machine learning algorithms for analysis.

Another challenge is the standardization and interoperability of IoT devices. With many manufacturers producing various sensors and devices, ensuring that these systems can communicate and share data effectively is crucial for seamless integration into supply chain operations.

1.5 Benefits of Track and Trace Supply Chain Systems

1.5.1 Improved Visibility and Transparency

Improved visibility and transparency in the track and trace supply chain are critically important in today's complex and globalized marketplace. With increasing consumer demands for accountability and the need for organizations to manage risk effectively, having a clear view of product journeys—from raw materials to end-users—is essential. Enhanced visibility allows stakeholders to monitor every aspect of the supply chain, including inventory levels, shipment locations, and product conditions, resulting in better decision-making and higher levels of operational efficiency.

One of the primary benefits of improved visibility in the supply chain is the ability to track products in real-time. Enhanced tracking capabilities, facilitated by advanced technologies such as RFID, GPS, and IoT sensors, enable businesses to know the precise location of their goods at any given moment. This real-time insight is invaluable for many industries, especially those dealing with perishable goods, pharmaceuticals, and high-value assets. For example, in the food industry, real-time tracking allows suppliers to ensure products are stored and transported under safe conditions, thus minimizing waste and ensuring strict compliance with health regulations. Similarly, for electronics and luxury goods, real-time tracking aids in preventing theft and counterfeiting while maintaining quality assurance throughout the supply chain.

1.5.2 Improved Efficiency and Productivity

Improved efficiency and productivity in the track and trace supply chain are pivotal for organizations aiming to enhance their operations and meet the ever-increasing demands of consumers. The ability to accurately track and trace products at every stage—from raw materials to final delivery—enables businesses to streamline processes, reduce waste, and optimize resource utilization. These improvements ultimately lead to cost savings, faster delivery times, and increased customer satisfaction.

One of the primary ways track and trace systems improve efficiency is by providing real-time visibility into inventory levels and product movements. Traditional inventory management methods often rely on periodic counts and manual records, which can lead to errors and delays. In contrast, modern track and trace technologies utilize IoT sensors, RFID tags, and automated data capture tools to provide continuous, real-time updates on stock levels and product locations. This accuracy allows businesses to quickly identify discrepancies, avoid overstocking or stockouts, and tailor their inventory practices to meet actual demand.

CHAPTER-2

LITERATURE SURVEY

2.1 Related Work

Udhayasanthiya; A. 1. P Yogananth; K. Priyadharshini; S. Mahalakshmi; R. Shilpasree; Customer demanding products in online shopping — A novel framework; 16-18 February 2017.

E-Commerce is the Digital Technology of buying or selling products online. In 2020, E-Commerce of India is predicted to bounce from certain billion of its income to tremendous increase in total income which is highest in the world. The sector of online purchase is prominent by the penetration of mobile, increasing internet, growing facility of e-payments among users has improved the purchase of e-commerce products in India. The branded products like televisions, mobile phones, laptops are the products that hits majorly on the e-commerce shopping, that are purchased by ordering through the mobile phones applications. On the other hand, the other pattern of online shopping is by bidding. Online bidding on Deal is exciting & easy. Buyers can bid or buy instantly, and sellers can use website to get rid of unwanted items. Hence the scope of the project is to develop an application for the User to shop by bidding their desired product online, then acquiring that product. The seller with satisfying bid rate can sell the product. The buyer can acquire bade product by delivering them to the buyers location.

[2]. Hafizur Rahman; Eshan Barua; Samanta Afrin; Ashikur Rahman; Mohammad Monirujjaman Khan; Bid & Buy: An Effective Online Based Platform for Client and Vendor; 08-10 April 2021.

This paper presents the design and the implementation of an effective online-based service providing platform for clients and vendors. It uses a very efficient way of buying and selling any products through an online bidding system. By using the system an organization such as government, business institution, or an individual can be more modular and organized in their buying and selling. The system has been designed in such a manner that the users of the system will get all the necessary information, and it is capable of providing an easy, effective and reliable way of buying and selling via an online bidding system which will change the E-commerce scenario of Bangladesh.

[3]. Vincentius Riandaru Prasetyo; Searching Cheapest Product on Three Different E-Commerce Using K-Means Algorithm; 30-31 August 2018.

The development of e-commerce is starting to change people's lifestyles, not least the people of Indonesia. The existence of e-commerce is very helpful user in buying and selling products. There are many e-commerce that can be found today. Some of the famous e-commerce in Indonesia are Bukalapak, Lazada, and Blibli. A large number of existing e-commerce makes users, especially buyers, have difficulty when looking for products at the cheapest price. This happens because each e-commerce offers different prices for the same product. This research aims to make the cheapest product search system in Bukalapak, Lazada, and Blibli using K-Means algorithm. The results of experiments showed that K-Means algorithm can be used to classify product data from Bukalapak, Lazada, and Blibli well. The results of the clustering process can also help for searching the cheapest products from the three e-commerce becomes faster. However, the number of clusters used will affect the effectiveness of the search process on the system.

[4] Thirumalai, S., & Sinha, K. K. (2005). Customer satisfaction with order fulfillment in retail supply chains: implications of product type in electronic B2C transactions. *Journal of Operations Management*, 23(3), 291-303

This paper focuses on the proverbial “last mile” of the retail supply chain – i.e., delivering products to the end-customer – and highlights the need for recognizing product type differences in configuring order fulfilment processes in electronic business-to-customer (B2C) transactions. The following two questions serve as the motivation for the study: Do customer expectations of order fulfilment processes vary across product types? Should the product type matter in configuring order fulfilment processes? From the studies in the marketing literature, we infer that customer satisfaction assessments are based on customer expectations of order fulfilment processes, and that these expectations systematically vary across the three product types: convenience goods (e.g., groceries, home and office supplies), shopping goods (e.g., ready-to-wear men, women, and kids’ apparel), and specialty goods (e.g., desktop and notebook computers, and wedding dresses). In particular, we posit that *ceteris paribus*, customer satisfaction with order fulfilment will decrease moving along a continuum of product types, from convenience goods to specialty goods. The empirical analysis for this study is based on data collected on dimensions of customer satisfaction with order fulfilment from a sample of 256 firms engaged in electronic B2C transactions. Firms included in our study

sample are such that their products can be classified into only one of three product types: convenience, shopping, or specialty goods. In essence, each firm in the study sample is a proxy for one of three product types. The results of the empirical analysis indicate that, on average, customers tend to have higher satisfaction levels with the order fulfilment process of convenience and shopping goods than with the order fulfilment process of specialty goods. We discuss the managerial implications of our results, contributions of the paper to the literature, limitations, and directions for future research.

[5] Ghadafi M. Razak , Linda C. Hendry and Mark Stevenson(2021). Supply chain traceability: a review of the benefits and its relationship with supply chain resilience. Received 25 February 2021Accepted 17 September 2021.

This paper adopts a systematic literature review approach to address these gaps, and thereby. To fully appreciate the benefits of traceability and its relationship with SCRes, it is first necessary to summarize the current state-of-the-art understanding of traceability systems in terms of: the drivers/motivations, the evolution of technology; and the challenges/barriers that inhibit its implementation. This is necessary given that there is a clear link between these issues and the benefits achieved. For example, the choice and subsequent success of a traceability system is affected by: the reasons for its adoption, the functionality of available technologies and how the SC partners overcome any barriers to adoption. Thus, understanding these issues enables a deeper understanding of the benefits. A comprehensive understanding of the benefits in turn enables the development of an understanding of the relationship between these benefits and SC Res. The criteria that determines whether a publication can provide information to answer the research questions was defined in advance of the actual literature search to avoid manipulating procedures based on the researcher's expectations. This SLR used a Microsoft Excel spreadsheet to record and monitor the data obtained from the selected studies. This cataloguing also provides an audit trail to map the claims made in the SLR to the source of the evidence. Scopus was the main search engine for this study because it is an up-to-date database of relevant journals. This was supplemented with the One Search Library, which also incorporates a range of major business and management databases such as EBSCOhost, SpringerLink, Science Direct, IEEE Xplore, Emerald Insight and Wiley Online Library. Despite the sophistication of each search engine, combining them widened the coverage and improved the search results.

[6] Moayad Al-Talib, Walid Al-Saad, Anan Alzoubi, Anthony I. Anosike: A systematic review of the literature on the use of information technologies in supply chain management.30 July 2024.

This paper aims to explore the potential benefits and challenges of applying various Information Technology (IT) tools, such as the Internet of Things (IoT), Big Data, information, and communications technology (ICT), Artificial Intelligence (AI), Digital Twin, Blockchain and information sharing, in different supply chain domains. While IT has been widely recognized as a key enabler of supply chain management, the existing literature has not sufficiently examined how these specific IT tools can be effectively integrated and utilised in supply chain contexts. Therefore, this paper contributes to the literature by providing a comprehensive review of the current state-of-the-art and identifying the research gaps and opportunities for future studies. Supply chains are dynamic systems with numerous moving parts, and it can be challenging to keep track of and understand all the relationships and interactions between different components ([Ramos et al., 2021](#)). AI and digital twins can assist us in developing more accurate and comprehensive supply chain models, which can then be used to enhance our understanding of their functionality and optimise them ([Dubey et al., 2022](#); [Badakhshan and Ball, 2023](#)).

[7] Martijn Koot, Martijn R.K. Mes, Maria E. Iacob: A systematic literature review of supply chain decision making supported by the Internet of Things and Big Data Analytics. 13 February 2021.

This paper aims at delivering an objective overview of the state-of-the-art IoT developments in today's SCM and logistics research. The main goal is to search for academic literature that explains how organizations can incorporate real-time data of physically operating objects into their decision making. Better understanding of the IoT's analytical capabilities stimulates future SCM research to customize information systems by proactively acting on the dynamic and stochastic nature of supply chains. Therefore, we have to map which type of IoT devices and analytical models are prescribed by scientists to improve supply chain performances. We conduct a systematic literature review (SLR) to explore how the real-time data of physically operating objects is applied into SCM and logistics research. The contribution of this research is twofold. First, we explain how, where, and why organizations could apply IoT devices into their SCM and logistics operations by conducting an integrated review towards the gathering, processing, and application of real-time data. Second, by using a proper classification of the state-of-the-art IoT developments.

[8] Sonia CisnerosCabrera , Grigory Pishchulov , Pedro Sampaio , Nikolay Mehandjiev , Zixu Liu , Sophia Kununka: An approach and decision support tool for forming Industry 4.0 supply chain collaborations. 29 January 2021.

The Industry 4.0 paradigm is changing the companies' focus towards organising production processes around the principles of interoperability between physical and cyber systems, decentralisation, real-time data analytics, service orientation, and modularity — which shall enable digital integration across the entire value chain, self-adaptation of production systems, and agile response to customer demand. Yet as large companies have already embraced the concept of Industry 4.0, its adoption by small and medium-sized enterprises (SMEs) is facing several challenges, such as resource constraints and lack of awareness of advanced technologies. Earlier work on decision support for team formation has been developed considering partner selection based on both objective and subjective criteria.

[9] Carolina Belotti Pedroso ^a, Wendy Lea Tate ^b, Andrea Lago da Silva ^c, Luiz César Ribeiro Carpinetti: Supplier development adoption: A conceptual model for triple bottom line (TBL) outcomes-1 July 2021.

The literature on supplier development has traditionally explored the economic dimension of sustainability and demonstrated it as an effective strategy to improve suppliers' results (Zhang et al., 2017). More recently, the environmental and the social elements of sustainability have also been incorporated in supplier development efforts, aiming to improve suppliers' environmental and social results (Trapp and Sarkis, 2016; Yawar and Seuring, 2017). Sustainable supplier development is a relatively new concept, particularly when taking a TBL focus. Considering the recent concerns from customers and other stakeholders demanding TBL results, the suppliers' performance regarding sustainability, and the potential that TBL supplier development offers to improve suppliers' sustainable results this research aims to investigate the primary strategies to improve suppliers' sustainable outcomes under supplier development context. This research differs from existing literature reviews as it synthesizes the existing literature about TBL supplier development, illustrating the primary practices that can be adopted and bringing a framework that can support improving suppliers' sustainable results.

CHAPTER-3

RESEARCH GAPS OF EXISTING METHODS

One major gap is the lack of standardization in data formats and communication protocols across different supply chain stakeholders. Existing methods often rely on proprietary systems that are incompatible with one another, leading to data silos and inefficiencies. This lack of interoperability makes it challenging to achieve a unified, end-to-end view of the supply chain, particularly in global operations involving multiple players with varying technological capabilities. Another limitation is the high cost and complexity of implementing advanced technologies, such as blockchain and IoT, particularly for small and medium-sized enterprises (SMEs). Many existing solutions are designed with large-scale operations in mind, neglecting the affordability and accessibility needs of smaller businesses. This creates a gap in adoption and leaves segments of the supply chain without adequate track-and-trace capabilities.

3.1 SYSTEM ANALYSIS

3.1.1 Existing System

Traditional supply markets typically rely on centralized databases and intermediaries, resulting in limited transparency and higher costs. This approach lacks the traceability and security required for modern supply chain systems. Additionally, these methods make it difficult to monitor product authenticity and track each transaction across the chain.

Disadvantages of existing systems

- Limited visibility and lack of centralized data access.
- Increased risk of counterfeit products and data tampering.
- High operational costs due to reliance on intermediaries.
- Delays in order processing and inventory management

3.1.2 Proposed System

The proposed system leverages blockchain technology to create a decentralized supply market platform. Users can browse items, add them to their cart, purchase products, and access purchase history. Admins have control over inventory management, user data, and transaction tracking. Smart contracts facilitate automatic transaction execution, eliminating the need for intermediaries and ensuring transparent, secure operations.

Advantages of Proposed System

- Real-time transaction visibility across the network.

- Blockchain's immutable ledger prevents data tampering.
- Reduced reliance on intermediaries lowers costs.
- Enhanced product traceability from origin to end-user.

3.1.3 Project flow:

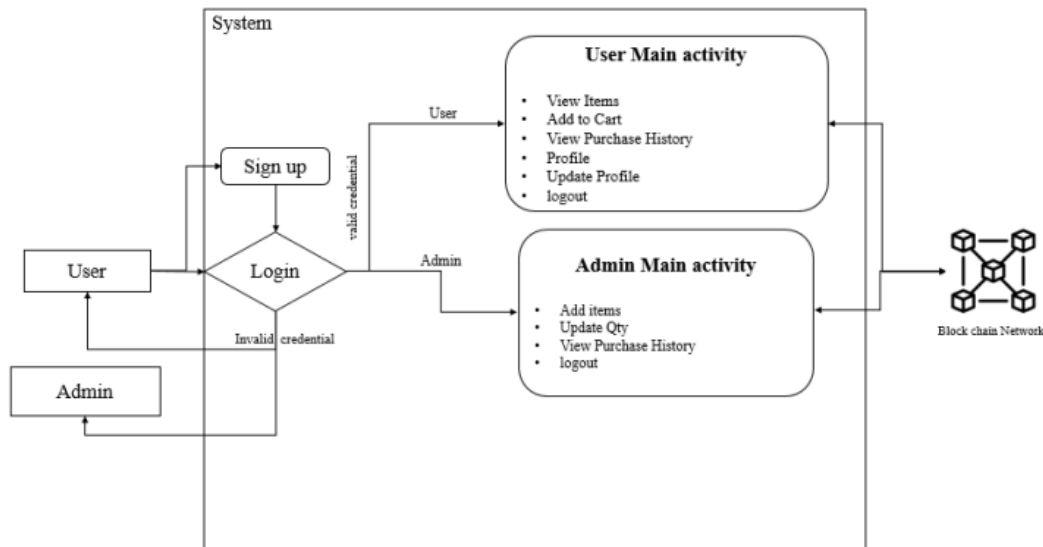


Figure 3.1.3

3.2 REQUIREMENT ANALYSIS:

3.2.1 Function and non-functional requirements

Requirement's analysis is very critical process that enables the success of a system or software project to be assessed. Requirements are generally split into two types: Functional and non-functional requirements.

Functional Requirements: These are the requirements that the end user specifically demands as basic facilities that the system should offer. All these functionalities need to be necessarily incorporated into the system as a part of the contract. These are represented or stated in the form of input to be given to the system, the operation performed and the output expected. They are basically the requirements stated by the user which one can see directly in the final product, unlike the non-functional requirements.

Examples of functional requirements:

- 1) Authentication of user whenever he/she logs into the system
- 2) System shutdown in case of a cyber-attack
- 3) A verification email is sent to user whenever he/she register for the first time on some software system.

Non-functional requirements: These are basically the quality constraints that the system must satisfy according to the project contract. The priority or extent to which these factors are implemented varies from one project to other. They are also called non-behavioral requirements.

They basically deal with issues like:

- Portability
- Security
- Maintainability
- Reliability
- Scalability
- Performance
- Reusability
- Flexibility

Examples of non-functional requirements:

- 1) Emails should be sent with a latency of no greater than 12 hours from such an activity.
- 2) The processing of each request should be done within 10 seconds
- 3) The site should load in 3 seconds whenever of simultaneous users are > 10000

3.2.2 Hardware Requirements

Processor	- I3/Intel Processor
Hard Disk	- 160GB
Key Board	- Standard Windows Keyboard
Mouse	- Two or Three Button Mouse
Monitor	- SVGA
RAM	- 8GB

3.2..3 Software Requirements

- Operating System : Windows 7/8/10
- Programming Language : Python, Solidity
- Libraries : Django, web3, Truffle, Ganache
- Front End : HTML, CSS, JavaScript,
- IDE/Workbench : Visual Studio Code.

CHAPTER-4

PROPOSED METHODOLOGY

The methodology for the blockchain-based supply market management platform integrates decentralized blockchain technology, smart contracts, and cryptographic hashing with Keccak to ensure secure and efficient supply chain management.

4.1 Smart Contracts:

Smart contracts form the core of the platform's automation. These self-executing contracts encode business logic and predefined rules for transactions, such as inventory updates, order fulfillment, and payment processing. When the conditions specified in a smart contract are met, the contract executes automatically, reducing the need for intermediaries, minimizing costs, and decreasing transaction delays. For example, upon confirming a payment, a smart contract can trigger inventory updates and initiate shipping procedures.

One key benefit of using smart contracts in a track-and-trace system is real-time visibility. Blockchain's immutable ledger ensures that every transaction, from raw material sourcing to final delivery, is recorded securely and can be accessed by authorized parties. For instance, when goods move through various checkpoints in the supply chain, a smart contract can automatically update the blockchain with relevant details such as location, time, and quality checks. This real-time tracking capability enhances accountability and prevents counterfeiting, a significant issue in industries like pharmaceuticals and luxury goods.

4.2 Keccak Hashing:

The platform uses the Keccak hashing algorithm (the basis of SHA-3) to enhance data security and integrity. Keccak generates a unique cryptographic hash for every transaction, which is stored on the blockchain. This ensures that transaction data is tamper-proof, as any alteration would produce a different hash, making it detectable. Keccak's robust hashing mechanism adds an extra layer of security to the decentralized system, ensuring that records remain immutable and traceable. User functionalities include registration, logging in, adding items to a cart, and managing profiles, while admins manage inventory, monitor activities, and oversee transaction history. Future enhancements could involve integrating IoT for real-time monitoring and smart contracts for predictive analytics and compliance.

The primary advantage of Keccak in track-and-trace systems lies in its robust resistance to collision attacks, where two different inputs produce the same hash. This property

ensures that each supply chain transaction has a unique identifier, making it virtually impossible for malicious actors to alter or forge records without detection. For example, when a product moves from a manufacturer to a distributor, its shipment details can be hashed using Keccak and stored on a blockchain. If someone attempts to manipulate this data, the hash will no longer match the original, immediately signaling a discrepancy.

4.3 Identification and Data Capture:

Identification and data capture are crucial components of a robust track and trace supply chain system. The primary objective is to improve transparency, ensure product authenticity, and enhance overall operational efficiency. The process begins with the identification of products at various stages of the supply chain, from manufacturing to distribution and retail. This typically involves the use of unique identifiers such as barcodes, QR codes, or RFID tags, which allow each item to be individually tracked and traced throughout its lifecycle. These identifiers enable businesses to monitor the movement of goods in real time, thereby reducing the chances of errors or fraud. The integration of identification and data capture systems in the supply chain creates a comprehensive and transparent framework.

4.4 Sustainability Tracking:

Sustainability tracking within a track and trace supply chain is an increasingly vital element for businesses aiming to minimize their environmental impact while meeting regulatory compliance and consumer expectations. The integration of sustainability metrics into tracking systems enables companies to monitor the ecological footprint of their products at each stage of the supply chain, from raw material sourcing to manufacturing, distribution, and disposal. By utilizing tools such as life cycle assessments (LCA) and carbon footprint calculations, businesses can collect data regarding resource consumption, emissions, and waste generation. This holistic view allows for informed decision-making about processes that can be optimized for sustainability.

4.5 Enhanced Customer Experience:

Enhanced customer experience in a track and trace supply chain is increasingly becoming a competitive differentiator for businesses across various industries. By implementing robust track and trace systems, companies can provide customers with real-time visibility into the status and location of their products throughout the supply chain. This transparency not only builds trust but also empowers customers by keeping them informed about their orders.

4.6 Real-Time Notifications and Alerts:

Real-time notifications and alerts are essential components of an effective track and trace supply chain, enhancing communication and transparency between businesses and their customers. In today's fast-paced market, immediate access to information is crucial for maintaining efficiency and customer satisfaction. By implementing real-time notification systems, companies can promptly inform customers about key milestones in their order's journey, such as processing updates, shipment confirmations, and delivery alerts. These proactive communications reduce uncertainty for consumers, helping them manage their expectations and enabling them to plan accordingly.

In addition to improving customer experiences, real-time notifications can optimize internal operations. By integrating advanced technologies such as IoT devices and GPS tracking, businesses can monitor assets and inventory in real-time. This capability allows companies to receive alerts about critical conditions, such as temperature fluctuations for sensitive products, unauthorized access to facilities, or equipment malfunctions. By responding quickly to these alerts, businesses can take corrective actions to prevent product spoilage, reduce operational costs, and improve overall supply chain integrity.

4.7 Product authentication:

Product authentication plays a crucial role in enhancing the integrity of the track and trace supply chain, addressing the challenges posed by counterfeit goods and fraudulent practices that threaten brand reputation and consumer safety. In industries ranging from luxury goods to pharmaceuticals and electronics, ensuring that products are genuine is paramount. To achieve this, organizations typically employ unique identifiers such as barcodes, QR codes, or embedded RFID tags that enable real-time tracking and verification at every stage of the supply chain. This not only allows businesses to confirm the authenticity of their products but also provides consumers with reassurance that they are purchasing legitimate items.

The advent of technologies like blockchain has revolutionized product authentication by enabling secure and transparent tracking of products from their origin to the end consumer. Blockchain's decentralized ledger system creates an immutable record of each transaction associated with a product, documenting its journey through manufacturing, distribution, and retail. This level of traceability ensures that every stakeholder in the supply chain—from manufacturers to consumers—can access verified information about a product's authenticity. By scanning a QR code, for instance, a consumer can see the entire history of the product, including where it was produced, its supply chain path, and even certifications of authenticity.

CHAPTER-5

OBJECTIVES

The primary objective of developing a track-and-trace supply chain platform is to achieve end-to-end visibility and transparency across all stages of the supply chain, from raw material sourcing to final delivery. The platform aims to address the challenges of data silos and lack of standardization by enabling interoperability through standardized data formats and communication protocols. It seeks to ensure data security and integrity by leveraging blockchain or decentralized methods while maintaining scalability and energy efficiency.

5.1 Achieve End-to-End Supply Chain Visibility:

One of the key benefits of achieving end-to-end visibility is the ability to proactively identify and address potential issues before they escalate. For instance, with real-time tracking, companies can monitor the location and condition of products throughout their journey. If a shipment is delayed or an anomaly is detected—such as temperature fluctuations in sensitive cargo—stakeholders can be alerted immediately, enabling timely interventions to mitigate risks. Such proactive management reduces the likelihood of costly delays, enhances customer satisfaction, and boosts the overall reliability of the supply chain.

5.2 Integrate Cost-Effective Solutions for SMEs:

Integrating cost-effective solutions for track and trace supply chain management is essential for small and medium-sized enterprises (SMEs) that seek to enhance their operational efficiency without incurring prohibitive expenses. Unlike larger corporations, which may have extensive resources to invest in sophisticated technology and infrastructure, SMEs often operate on tighter budgets, making it imperative for them to adopt scalable and affordable solutions. Fortunately, advancements in technology have democratized access to effective track and trace systems, enabling SMEs to leverage tools that enhance visibility, improve inventory management, and streamline logistics without breaking the bank.

5.3 Automate Processes with Smart Contracts:

Automating processes with smart contracts represents a transformative opportunity for enhancing the efficiency and accuracy of track and trace supply chains. Smart contracts, which are self-executing agreements with the terms of the contract directly written into code on a blockchain, streamline various supply chain operations by eliminating the need for intermediaries and reducing transaction times. This technology automates compliance,

payment processes, and record-keeping, allowing participants in the supply chain to focus on core activities rather than being bogged down by administrative tasks.

5.4 Enhance Risk Management and Anomaly Detection:

Enhancing risk management and anomaly detection within track and trace supply chains is essential for maintaining operational resilience and ensuring product integrity. As supply chains become increasingly complex and interconnected, the potential for disruptions—such as delays, fraud, or quality control issues—also escalates. Implementing advanced technologies and analytical methods can empower organizations to proactively identify and mitigate these risks, ultimately safeguarding their operations and maintaining customer trust.

Training employees to recognize and report unusual occurrences—not only in terms of operational anomalies but also regarding compliance and ethical standards—empowers frontline staff to act as the first line of defense against potential risks. An engaged workforce that understands the importance of risk management is better equipped to respond promptly to emerging issues, fostering a proactive mindset that permeates the entire supply chain.

5.5 Facilitate Regulatory Compliance:

Facilitating regulatory compliance within track and trace supply chains is paramount for businesses operating in regulated industries such as pharmaceuticals, food and beverage, automotive, and electronics. Regulatory bodies often impose stringent requirements on how products are tracked and traced throughout the supply chain to ensure quality, safety, and accountability. Adhering to these regulations not only helps organizations avoid legal penalties but also enhances their reputation and fosters consumer trust. Therefore, implementing effective track and trace solutions is vital for compliance management.

The decentralized and immutable nature of blockchain ensures that data related to product movement and handling is securely recorded and can't be altered retroactively. This transparency is crucial for industries like food and pharmaceuticals, where maintaining detailed histories of product origins and handling is essential for public health and safety. In the event of a product recall or trace-back, organizations can quickly access the complete and tamper-proof history of a product, allowing for rapid response and minimizing risks to consumers. Collaboration with supply chain partners also plays a crucial role in meeting regulatory standards. Organizations need to establish clear communication channels and expectations regarding compliance with their suppliers, manufacturers, and logistics providers.

CHAPTER-6

SYSTEM DESIGN & IMPLEMENTATION

6.1 Introduction of Input design

Input design is a crucial phase in system development that focuses on how data is entered into a system. It ensures data accuracy, efficiency, and user-friendliness by defining formats, layouts, and validation rules for inputs. Effective input design minimizes errors and facilitates smooth user interactions with the system, making data entry intuitive and streamlined. By structuring the data input process appropriately, it helps enhance overall system performance and ensures high-quality data is consistently captured for processing and analysis.

6.2 Output Design

Output design is critical in this project, as it determines how the system communicates the results of encryption, watermark verification, and decryption to the receiver. The system's output includes the decrypted message, tamper verification results, and reports on the integrity of the watermark.

In the context of "A Novel Hybrid Image Synthesis-Mapping Framework for Steganography without Embedding", output design must ensure that the receiver is provided with clear and accurate information, including the decrypted message or any tampering alerts. The output screens should be user-friendly, providing all necessary information in a straightforward format, while also safeguarding sensitive information.

Objectives of Output Design:

- **Serve the Intended Purpose:** The output should meet the needs of the system, delivering decrypted messages securely or providing tamper-detection reports if the image integrity is compromised.
- **Meet End-User Requirements:** The output design must ensure that both senders and receivers understand the results. For example, if a mismatch occurs in watermark verification, the system should clearly display the mismatch percentage and alert the user about possible tampering.
- **Deliver the Appropriate Quantity of Output:** Only the necessary information should be

shown, such as the decrypted message, percentage of watermark mismatch, or a tampering report.

- **Format the Output Properly:** The decrypted message, watermark verification results, or tamper alerts should be displayed in a clear, concise, and understandable format. Graphical indicators or reports (e.g., the percentage of tampering detected) can help users better understand the system's output.
- **Timeliness of Output:** Outputs must be provided in real-time, ensuring the receiver can make informed decisions about the integrity of the message without delay.

By maintaining these output design principles, the system will ensure clear communication between the sender and receiver, providing secure and tamper-evident output that meets the objectives of the project.

6.3 UML diagrams

UML stands for Unified Modelling Language. UML is a standardized general-purpose modelling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group. The goal is for UML to become a common language for creating models of object-oriented computer software. In its current form UML is comprised of two major components: a Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML.

The Unified Modelling Language is a standard language for specifying, Visualization, Constructing and documenting the artefacts of software system, as well as for business modelling and other non-software systems. The UML represents a collection of best engineering practices that have proven successful in the modelling of large and complex systems.

The UML is a very important part of developing objects-oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

6.3.1 USE CASE DIAGRAM

- ▶ A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis.
- ▶ Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies

between those use cases.

- The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.

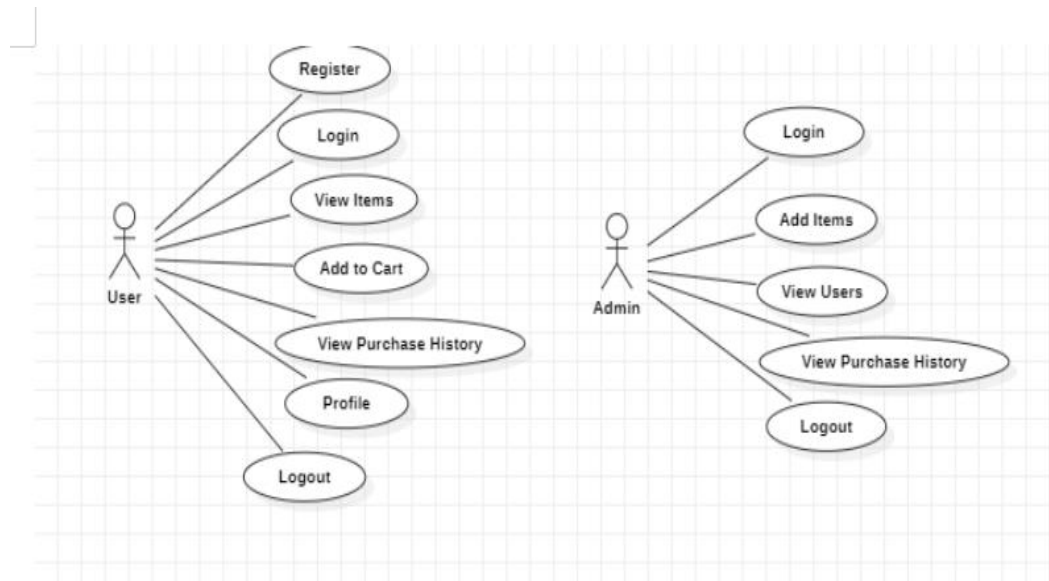


Figure 6.3.1

6.3.2 CLASS DIAGRAM

In software engineering, a class diagram in the Unified Modeling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among the classes. It explains which class contains information.

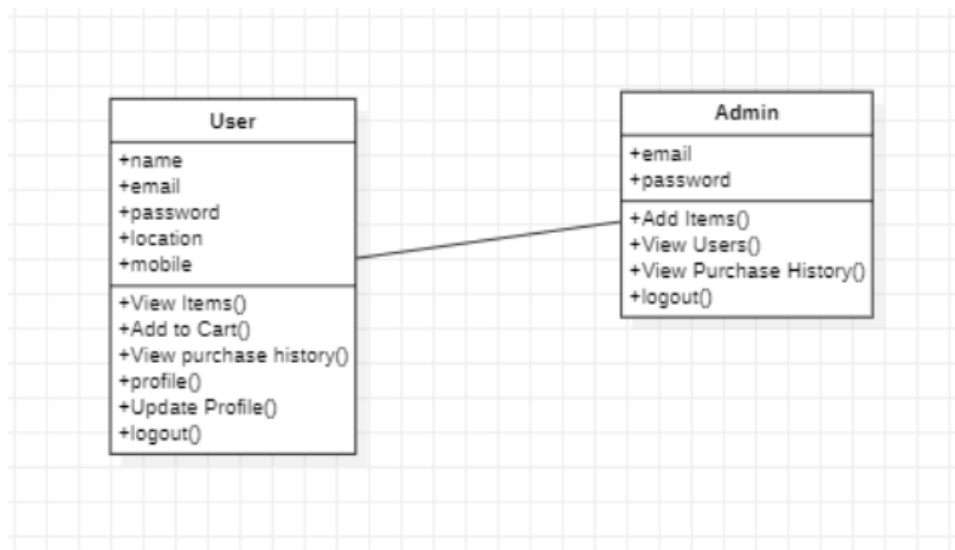


Figure 6.3.2

6.3.3 SEQUENCE DIAGRAM

- ▶ A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order.
- ▶ It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams.

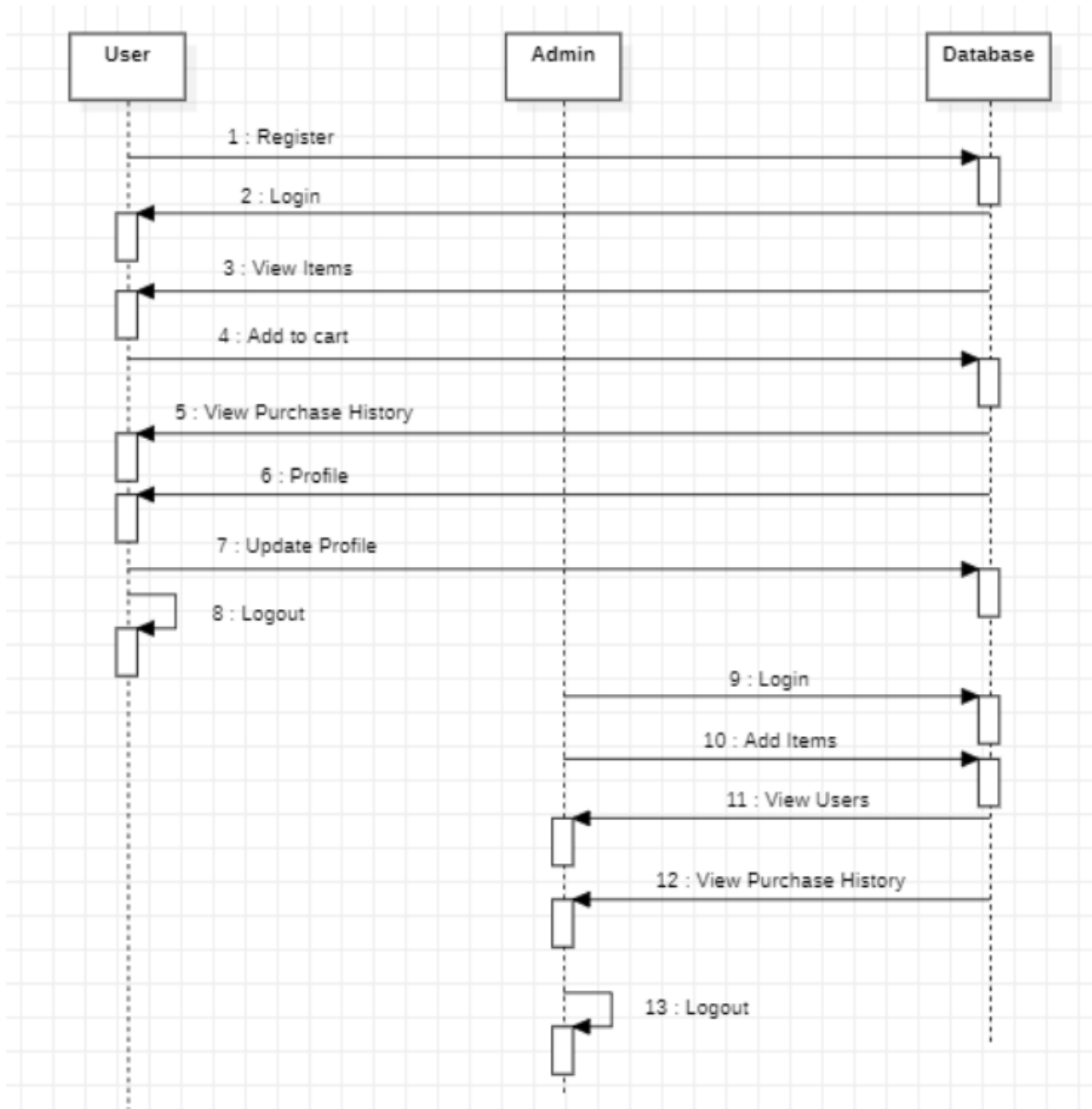


Figure 6.3.3

6.3.4 COLLABORATION DIAGRAM:

In collaboration diagram the method call sequence is indicated by some numbering technique as shown below. The number indicates how the methods are called one after another. We have taken the same order management system to describe the collaboration diagram. The method calls are similar to that of a sequence diagram. But the difference is that the sequence diagram does not describe the object organization whereas the collaboration diagram shows the object organization.

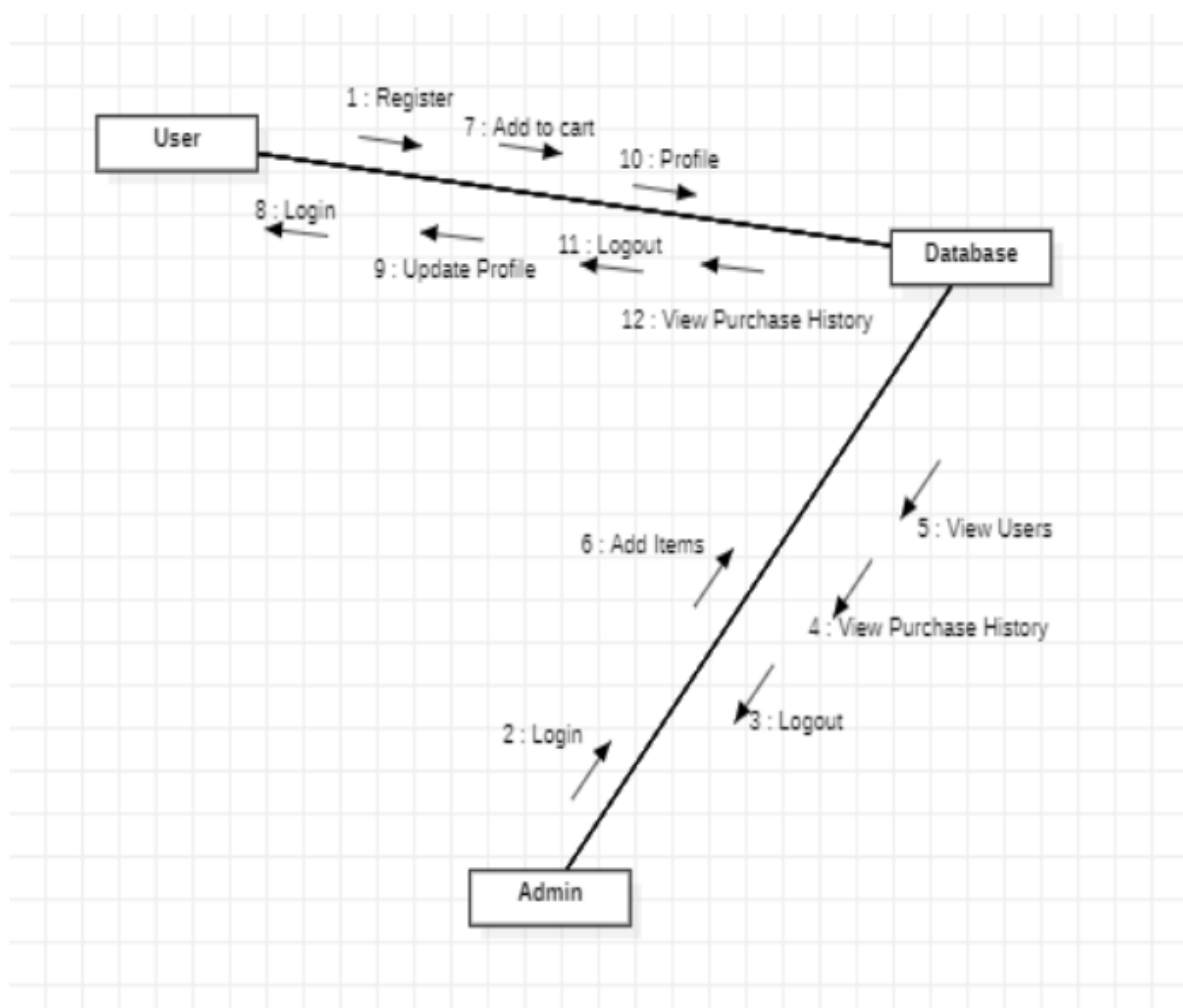


Figure 6.3.4

6.3.5 ACTIVITY DIAGRAM:

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the Unified Modelling Language, activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system. An activity diagram shows the overall flow of control.

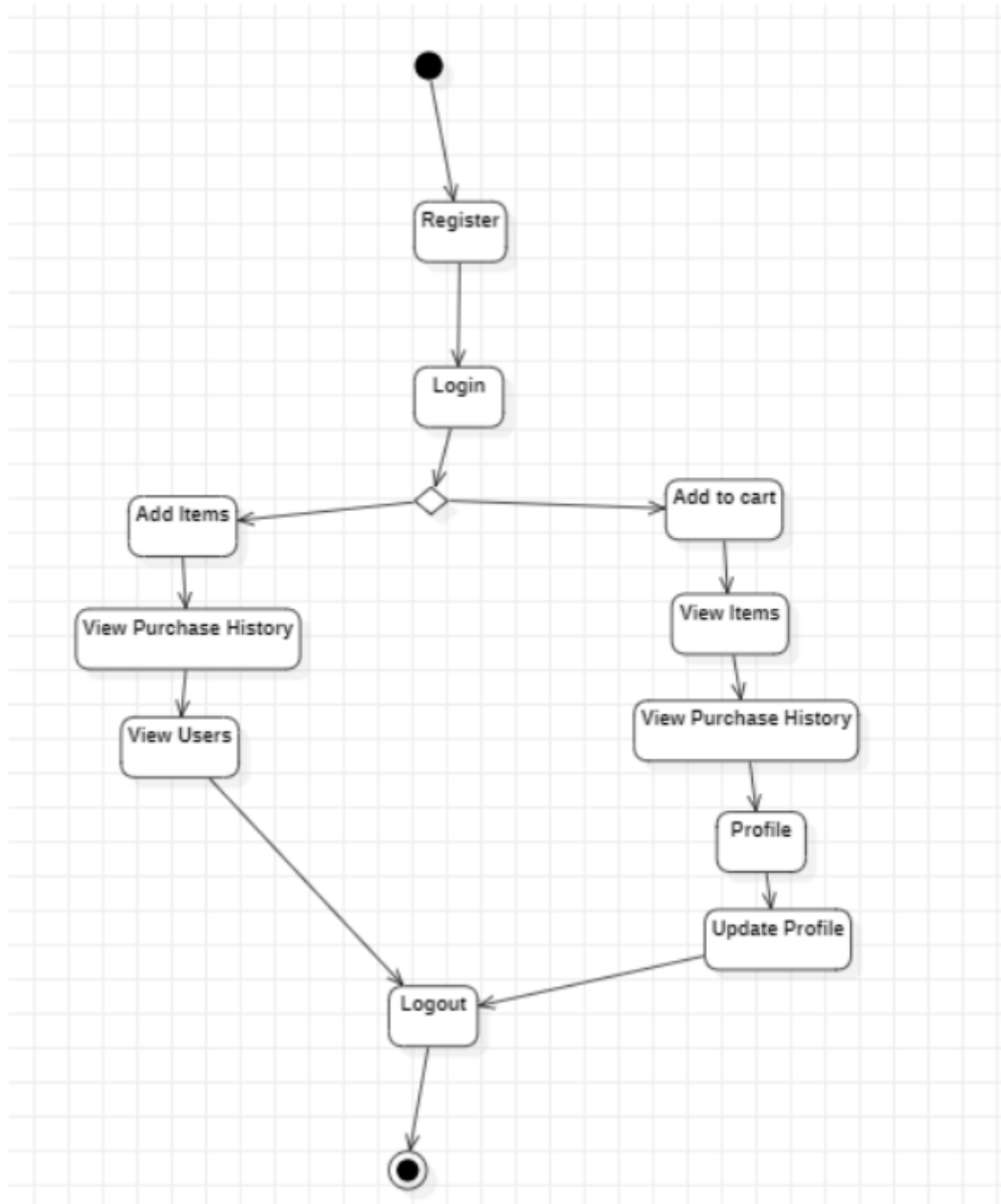


Figure 6.3.5

6.3.6 DEPLOYMENT DIAGRAM

Deployment diagram represents the deployment view of a system. It is related to the component diagram. Because the components are deployed using the deployment diagrams. A deployment diagram consists of nodes. Nodes are nothing but physical hardware's used to deploy the application.

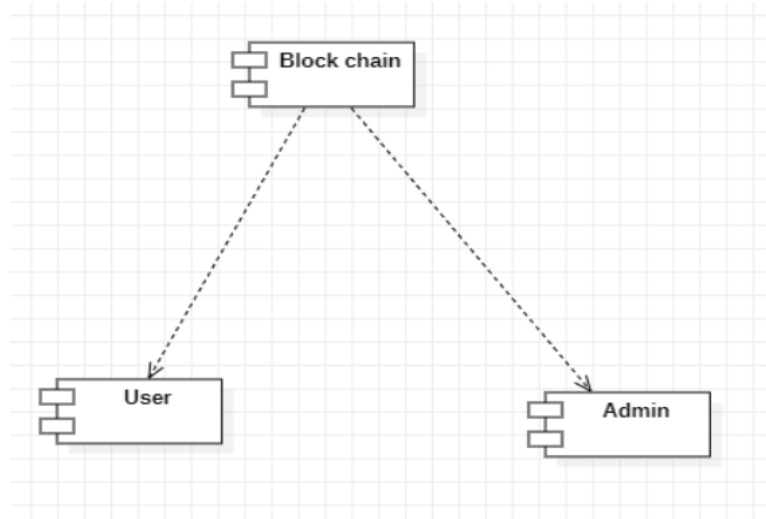


Figure 6.3.6

6.3.7 COMPONENT DIAGRAM:

A component diagram, also known as a UML component diagram, describes the organization and wiring of the physical components in a system. Component diagrams are often drawn to help model implementation details and double-check that every aspect of the system's required function is covered by planned development.

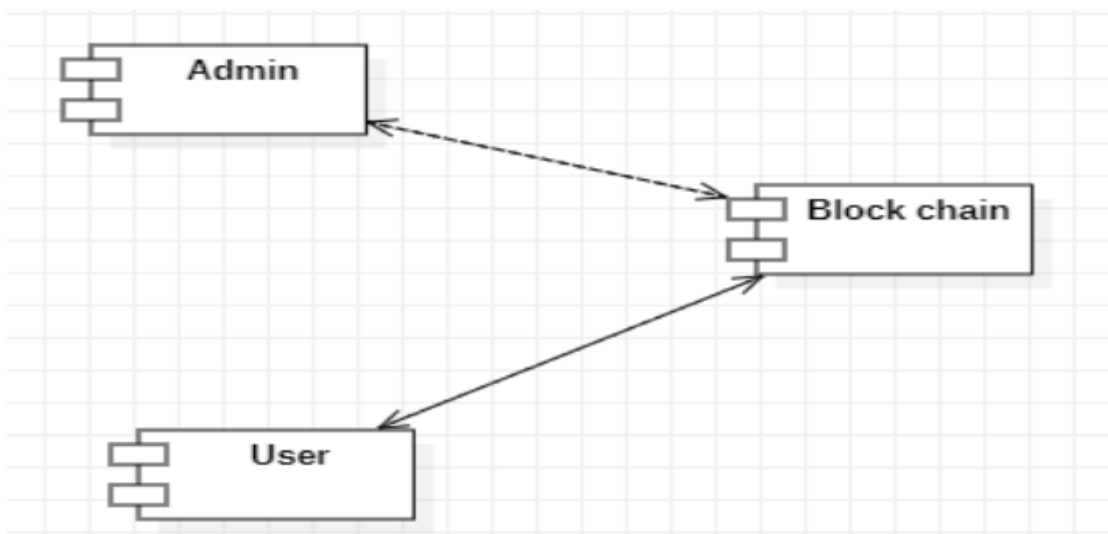


Figure 6.3.7

6.4 Data Flow diagrams

A Data Flow Diagram (DFD) is a traditional way to visualize the information flows within a system. A neat and clear DFD can depict a good amount of the system requirements graphically. It can be manual, automated, or a combination of both. It shows how information enters and leaves the system, what changes the information and where information is stored. The purpose of a DFD is to show the scope and boundaries of a system as a whole. It may be used as a communications tool between a systems analyst and any person who plays a part in the system that acts as the starting point for redesigning a system.

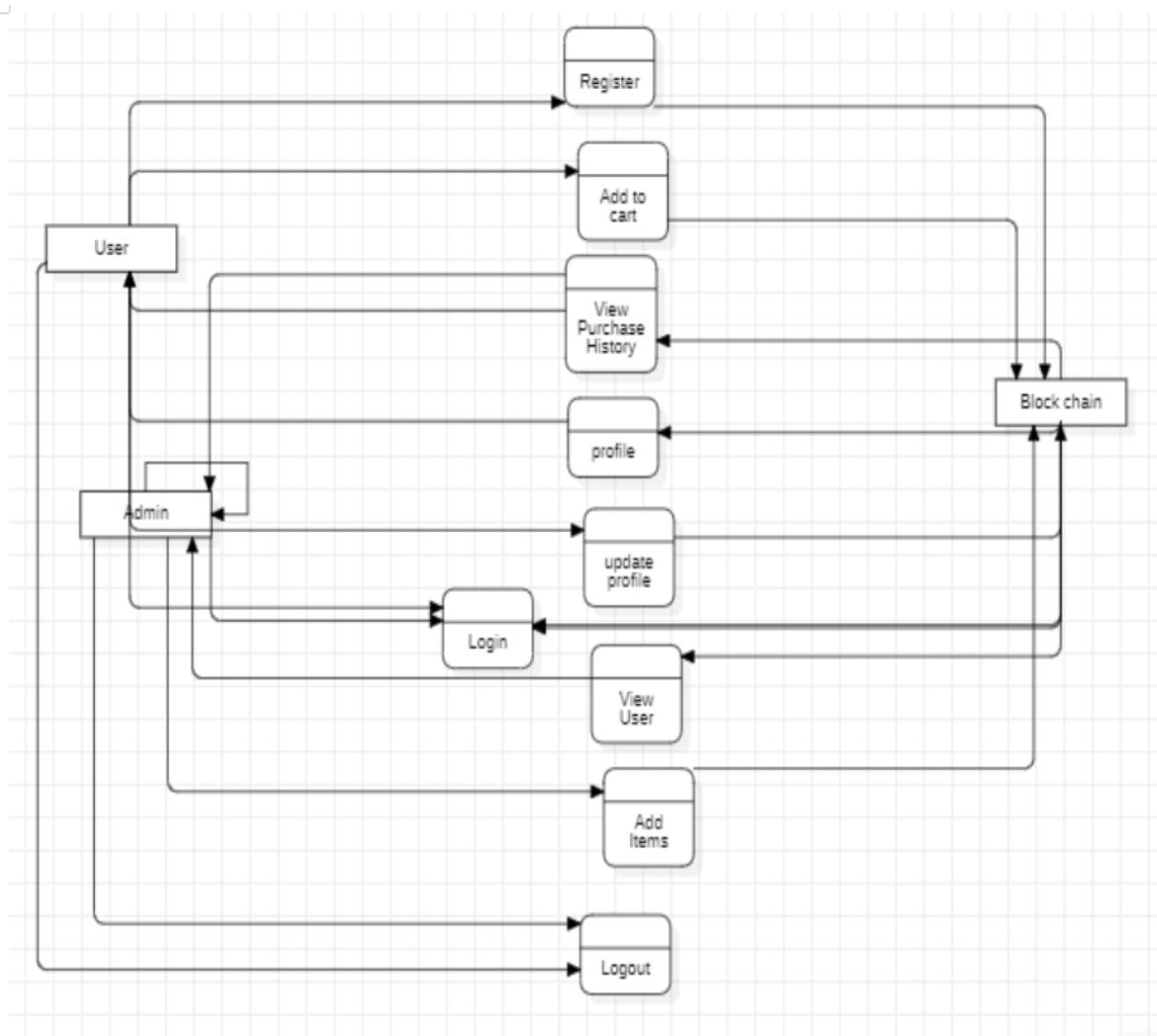


Figure 6.4

6.5 Architecture Diagram:

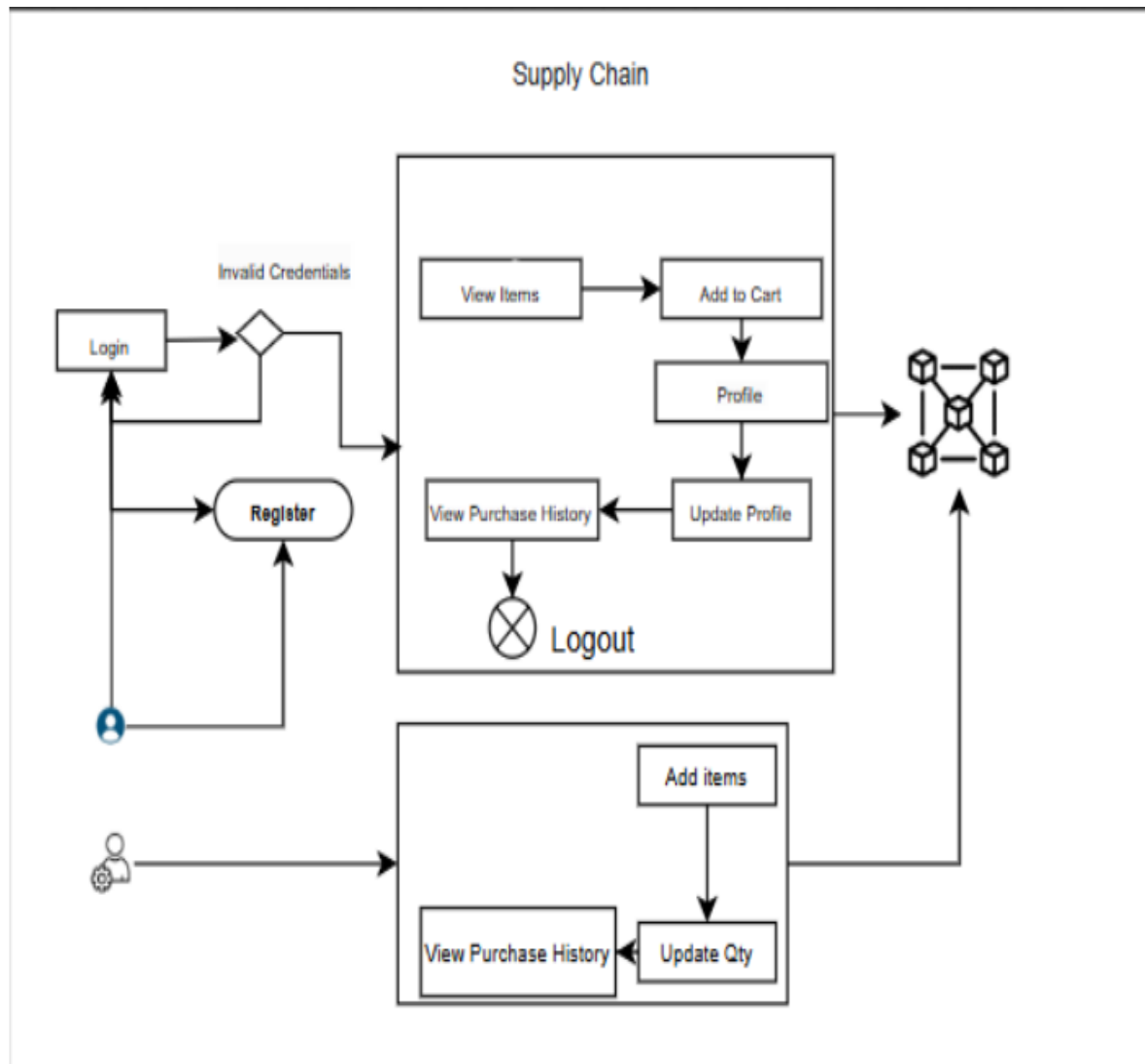
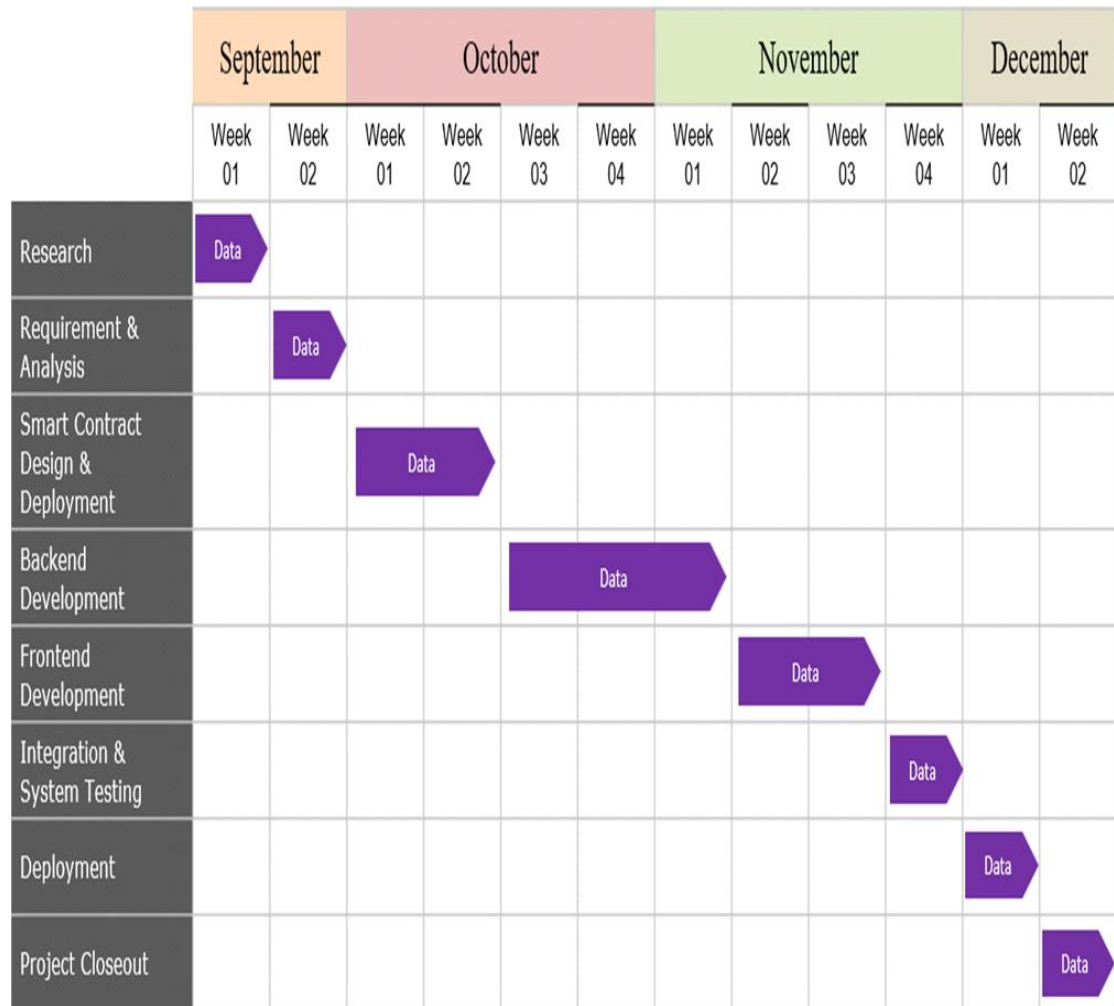


Figure 6.5

CHAPTER-7

TIMELINE FOR EXECUTION OF PROJECT (GANTT CHART)



CHAPTER-8

OUTCOMES

➤ **Enhanced Supply Chain Visibility:**

One of the primary outcomes is improved visibility throughout the supply chain, allowing businesses to monitor the movement of products in real time. This end-to-end transparency helps in quickly identifying any bottlenecks, delays, or disruptions, enabling proactive decision-making to maintain smooth operations.

➤ **Improved Product Traceability and Security:**

The Track and Trace system will enhance product traceability by providing a detailed record of each product's journey from manufacturing to the end customer. This increased level of traceability helps to ensure product authenticity, prevent counterfeiting, and reduce the risks associated with theft and fraud in the supply chain.

➤ **Optimized Inventory Management:**

An expected outcome is more accurate inventory management, with precise tracking of stock levels and product movements. Businesses will benefit from reduced instances of overstocking or stockouts, leading to optimized inventory levels, better demand forecasting, and efficient warehouse management.

➤ **Faster Response to Supply Chain Disruptions:**

With real-time tracking and data analysis capabilities, the Track and Trace system will enable businesses to respond swiftly to supply chain disruptions. Whether it's a delay in shipment, product recall, or quality issue, the system will provide actionable insights to minimize impact and implement corrective actions quickly.

➤ **Increased Customer Satisfaction:**

By providing reliable and transparent information to customers regarding their orders and shipment status, the Track and Trace system will significantly enhance customer satisfaction. Improved delivery times and accurate tracking details will lead to better customer experiences and stronger brand loyalty.

➤ **Compliance with Regulatory Standards:**

The Track and Trace system will help businesses comply with industry standards and regulatory requirements by maintaining accurate records and documentation of product movements. This compliance is crucial in sectors like pharmaceuticals, food and beverages, and electronics, where traceability and safety standards are stringent.

➤ **Cost Reduction and Increased Efficiency:**

The implementation of a Track and Trace system will lead to reduced operational costs through optimized logistics, reduced manual interventions, and more efficient supply chain processes. By minimizing errors and delays, businesses can improve productivity and overall supply chain performance.

➤ **Data-Driven Decision Making:**

The system will generate a wealth of data that can be analysed to gain insights into supply chain trends, performance metrics, and areas for improvement. These data-driven insights will empower businesses to make informed decisions, optimize supply chain strategies, and identify new opportunities for growth.

➤ **Environmental Sustainability:**

One of the expected outcomes is the promotion of environmentally friendly supply chain practices. By optimizing transportation routes, reducing unnecessary logistics movements, and improving inventory management, the Track and Trace system will contribute to a lower carbon footprint and more sustainable operations.

➤ **Competitive Advantage:**

The successful implementation of a Track and Trace system will provide businesses with a competitive advantage by enabling them to deliver higher levels of efficiency, transparency, and reliability in their supply chain operations. This differentiation in the market can help attract more customers and increase market share.

CHAPTER-9

RESULTS AND DISCUSSIONS

User Module:

Users register, log in, and manage their accounts securely. They can add items to their cart, make purchases, and view their transaction history for easy tracking. Profile updates allow users to maintain accurate account information.

At the core of the user module is a dashboard that provides users with a comprehensive overview of the supply chain activities relevant to their role. For instance, manufacturers can view data on raw material sources, production schedules, and quality control inspections. Suppliers can track inventory levels and order statuses, while logistics providers can monitor shipment routes and delivery confirmations. A well-designed dashboard presents real-time data in a visually appealing and easily digestible format, utilizing charts, graphs, and alerts to highlight important trends or anomalies. This level of visibility allows users to make informed decisions quickly and enhances their ability to respond to potential issues proactively.

Another important aspect of the user module is its reporting and analytics capabilities. Users should have access to customizable reporting tools that allow them to generate insights based on the data collected within the supply chain. For example, users can analyze trends in product movement, identify bottlenecks, and assess supplier performance over time. These analytics can inform strategic decision-making and operational improvements, enabling users to optimize their supply chain processes and enhance overall efficiency.

Admin Module:

Admins use secure login credentials to access the platform. They manage inventory by adding or updating items, overseeing stock levels, and monitoring user activity. Admins also have access to user management and transaction tracking, providing insights into purchasing trends and user interactions. This module provides the tools necessary for managing user accounts, configurations, and data integrity, ensuring that the system operates smoothly and adheres to organizational and regulatory standards. By centralizing administrative functions, this module not only enhances operational efficiency but also establishes a solid foundation for maintaining the system's accountability and effectiveness.

One of the primary responsibilities of the admin module is user management. Administrators can create, modify, and deactivate user accounts, as well as define roles and permissions based on organizational needs. This role-based access control ensures that each user operates within defined boundaries, allowing them to access only the information

necessary for their respective functions. By using the admin module to enforce security protocols, organizations can safeguard sensitive data and reduce the risk of unauthorized access or data breaches, thereby fostering a more secure supply chain environment.

Data integrity and management are essential functions within the admin module. Administrators are responsible for maintaining the accuracy and reliability of the data collected throughout the supply chain. This includes implementing data validation rules, managing data imports and exports, and ensuring that historical data is properly archived and accessible for reporting and audits.

Compliance monitoring is another crucial aspect of the admin module. Administrators can establish and track compliance criteria and protocols tailored to industry regulations, such as FDA guidelines in pharmaceuticals or food safety standards. The module can be designed to generate alerts and notifications for upcoming compliance deadlines, facilitating timely actions that ensure adherence to regulatory obligations. Additionally, it can store and organize all compliance documentation, making it easily accessible for audits and inspections.

9.1 System Study:

In this phase, the feasibility of the proposed "Hybrid Image Synthesis-Mapping Framework for Steganography without Embedding" system is analyzed to ensure that it is viable, both technically and economically. The feasibility study focuses on three key areas: economic feasibility, technical feasibility, and social feasibility. These assessments ensure that the system is cost-effective, technically implementable, and socially accepted by its users.

9.1.1 Economic Feasibility

The blockchain-based supply market management platform is economically feasible, as it reduces transaction costs by eliminating intermediaries and enhancing process efficiency. Its decentralized framework lowers operational expenses through automated smart contracts. The platform's scalability allows for future integration of predictive analytics and IoT, optimizing inventory management and compliance, ultimately delivering long-term cost savings and value.

9.1.2 Technical Feasibility

The platform's technical feasibility is ensured by leveraging blockchain technology for secure, transparent transactions and smart contracts for automated processes. Its architecture supports decentralized data management, ensuring immutability and traceability. The system's modular design facilitates integration with future technologies like IoT and predictive analytics, while scalable infrastructure accommodates user growth and evolving supply chain requirements.

9.1.3 Social Feasibility

Social feasibility measures the acceptance of the system by its intended users. The project focuses on a user-friendly web interface, making it easy for both senders and receivers to navigate. Minimal training is required to use the system effectively, as the process of sending and receiving messages, embedding, and verifying watermarks is simple and intuitive. Users are likely to accept and engage with the system, as it enhances communication security without introducing significant complexity.

9.2 System Testing:

Testing is critical in ensuring that the system operates according to its requirements and functions as expected. Different testing methodologies are employed to uncover potential errors and verify that the system meets both functional and non-functional requirements. Below are the various types of testing applied to the proposed system:

9.2.1 Unit Testing

Unit testing involves validating individual components or functions of the blockchain-based platform to ensure each performs as expected. It focuses on testing smaller modules, such as user registration, login, cart management, and smart contract execution. By isolating these functions, unit tests can identify specific issues early in the development process, enhancing code quality. Automated testing tools like Mocha or Jest can be used to perform repeated tests efficiently, ensuring stability and accuracy. Thorough unit testing minimizes bugs and supports reliable system functionality.

9.2.2 Integration Testing

Integration testing verifies the interaction between various components of the blockchain-based platform, such as user management, transaction handling, and smart contracts. It ensures that these modules work seamlessly together, detecting issues that may arise from component interactions. By testing data flow and combined functionalities, integration testing ensures a cohesive, robust system, reducing the risk of failures in real-world usage.

9.2.3 Functional Testing

Functional testing ensures that the system meets all specified functional requirements. For this project, functional testing involves checking that the encryption, embedding, watermarking, and tamper detection processes work as expected. Examples of functional tests include verifying that the system correctly encrypts plaintext that the watermark can detect tampering, and that only intact images allow decryption.

9.2.4 White Box Testing

In white box testing, the internal workings of the system are known, and specific areas of the code are targeted for testing. This type of testing is performed on critical components like encryption algorithms and steganography functions to ensure they operate as intended and handle edge cases such as incomplete or corrupted inputs.

9.2.5 Black Box Testing

In black box testing, the system is tested without knowledge of the internal code. Testers provide input and verify that the system produces the correct outputs based on functional requirements. For example, testers might input a plaintext message, encrypt it, and then check whether the system correctly embeds and extracts the message without knowing the internal workings of the encryption or steganography processes.

9.3 Test cases:

Input	Output	Result
Input text	A watermark embedded image with encrypted text	Success

S.NO	Test cases	I/O	Expected O/T	Actual O/T	P/F
1	Register user	Enter username, email, password, address, mobile number	Registered successfully.	Registered successfully	P
2	Login user	Enter email, password.	Login successful	Login successful	P
3	Send Text	Text	A watermark embedded image with encrypted	A watermark embedded image with encrypted text	P

CHAPTER-10

CONCLUSION

In conclusion, The implementation of a Track and Trace supply chain system has become essential for businesses aiming to enhance transparency, efficiency, and security in their supply chain operations. As supply chains grow more complex and globalized, the need for robust tracking solutions that provide real-time visibility and data analytics has never been more critical. Through this report, we have explored the significance of Track and Trace systems, examined the enabling technologies, discussed their benefits and challenges, and outlined a structured methodology for their successful deployment.

Despite the numerous advantages of Track and Trace systems, such as improved inventory management, enhanced product security, and compliance with industry regulations, there are also challenges, including high implementation costs, data privacy concerns, and integration issues with existing systems. Addressing these limitations will require continuous innovation, investment in technology, and a collaborative effort from all stakeholders in the supply chain. The objectives of Track and Trace systems align with the growing demand for greater transparency and accountability in supply chain management. By adopting the right strategies and leveraging advanced technologies like RFID, IoT, GPS, and blockchain, businesses can not only optimize their operations but also build stronger relationships with their partners and customers through improved trust and reliability.

- Ensure Product Quality and Safety
- Boost Efficiency and Reduce Costs
- Build Trust and Reputation
- Promote Sustainability
- Leverage Data-Driven Insights

REFERENCES

- [1]. P. Dudczyk, J. K. Dunston and G. V. Crosby, "Blockchain Technology for Global Supply Chain Management: A Survey of Applications, Challenges, Opportunities and Implications," in *IEEE Access*, vol. 12, pp. 70065-70088, 2024, doi: 10.1109/ACCESS.2024.3399759.
keywords: {Blockchains; Supply chains; Supply chain management; Business; Surveys; Peer-to-peer computing; Smart contracts; Globalization; Blockchain; blockchain applications; blockchain technology; global supply chain management; literature review; supply chain management; survey},
- [2]. Beaulieu, Martin, and Omar Bentahar. "Digitalization of the healthcare supply chain: A roadmap to generate benefits and effectively support healthcare delivery." *Technological forecasting and social change* 167 (2021): 120717.
- [3]. Dubey, Rameshwar, et al. "Impact of artificial intelligence-driven big data analytics culture on agility and resilience in humanitarian supply chain: A practice-based view." *International Journal of Production Economics* 250 (2022): 108618.
- [4]. Wamba, Samuel Fosso, et al. "The performance effects of big data analytics and supply chain ambidexterity: The moderating effect of environmental dynamism." *International Journal of Production Economics* 222 (2020): 107498.
- [5]. Gupta, Himanshu, et al. "Strategies to overcome barriers to innovative digitalisation technologies for supply chain logistics resilience during pandemic." *Technology in Society* 69 (2022): 101970.
- [6]. Helo, Petri, and A. H. M. Shamsuzzoha. "Real-time supply chain—A blockchain architecture for project deliveries." *Robotics and Computer-Integrated Manufacturing* 63 (2020): 101909.
- [7]. Njomane, Linda, and Arnesh Telukdarie. "Impact of COVID-19 food supply chain: Comparing the use of IoT in three South African supermarkets." *Technology in Society* 71 (2022): 102051.
- [8]. Wang, Hui, et al. "Assessing sustainability performance of global supply chains: An input-output modelling approach." *European journal of operational research* 285.1 (2020): 393-404.
- [9]. Deng, Weisheng, et al. "Effects of supply chain competition on firms' product sustainability strategy." *Journal of Cleaner Production* 275 (2020): 124061.
- [10]. Feng, Huanhuan, et al. "Applying blockchain technology to improve agri-food traceability: A review of development methods, benefits and challenges." *Journal of cleaner production* 260 (2020): 121031.
- [11]. Rahman, Hafizur, et al. "Bid & Buy: An Effective Online Based Platform for Client and Vendor." 2021 5th International Conference on Computing Methodologies and Communication (ICCMC). IEEE, 2021.

APPENDIX-A

PSUEDOCODE

BEGIN TrackAndTraceSupplyChain

// Initialization

Initialize database for tracking items:

ItemsTable(ItemID, ProductName, BatchNumber, CurrentLocation, Status, Timestamp)

LocationsTable(LocationID, LocationName, LocationType)

EventLogs(EventID, ItemID, EventType, Location, Timestamp)

// Function: Register Item

FUNCTION RegisterItem(ItemID, ProductName, BatchNumber, InitialLocation):

Add new record to ItemsTable with:

ItemID = ItemID

ProductName = ProductName

BatchNumber = BatchNumber

CurrentLocation = InitialLocation

Status = "Registered"

Timestamp = Current Timestamp

Log event in EventLogs: "Item Registered"

END FUNCTION

// Function: Update Item Location

FUNCTION UpdateLocation(ItemID, NewLocation):

IF ItemID exists in ItemsTable:

Update CurrentLocation in ItemsTable with NewLocation

Update Timestamp in ItemsTable

Log event in EventLogs: "Location Updated"

ELSE:

Print "Item not found"

END FUNCTION

```
// Function: Update Item Status
FUNCTION UpdateStatus(ItemID, NewStatus):
    IF ItemID exists in ItemsTable:
        Update Status in ItemsTable with NewStatus
        Update Timestamp in ItemsTable
        Log event in EventLogs: "Status Updated"
    ELSE:
        Print "Item not found"
END FUNCTION

// Function: Query Item Details
FUNCTION QueryItem(ItemID):
    IF ItemID exists in ItemsTable:
        Retrieve and return details from ItemsTable
    ELSE:
        Print "Item not found"
END FUNCTION

// Function: Generate Trace Report
FUNCTION GenerateTraceReport(ItemID):
    IF ItemID exists in EventLogs:
        Retrieve all EventLogs for the given ItemID sorted by Timestamp
        Return trace history
    ELSE:
        Print "Item not found"
END FUNCTION

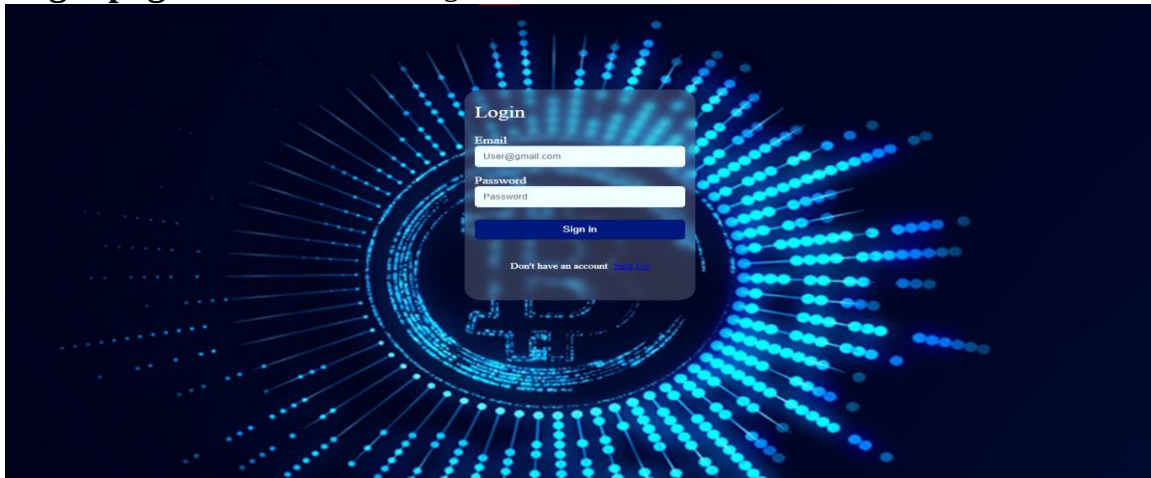
// Example Usage
RegisterItem("12345", "Laptop", "Batch001", "Factory A")
UpdateLocation("12345", "Warehouse B")
UpdateStatus("12345", "In Transit")
UpdateLocation("12345", "Retailer C")
QueryItem("12345")
GenerateTraceReport("12345")
```

```
END TrackAndTraceSupplyChain
```

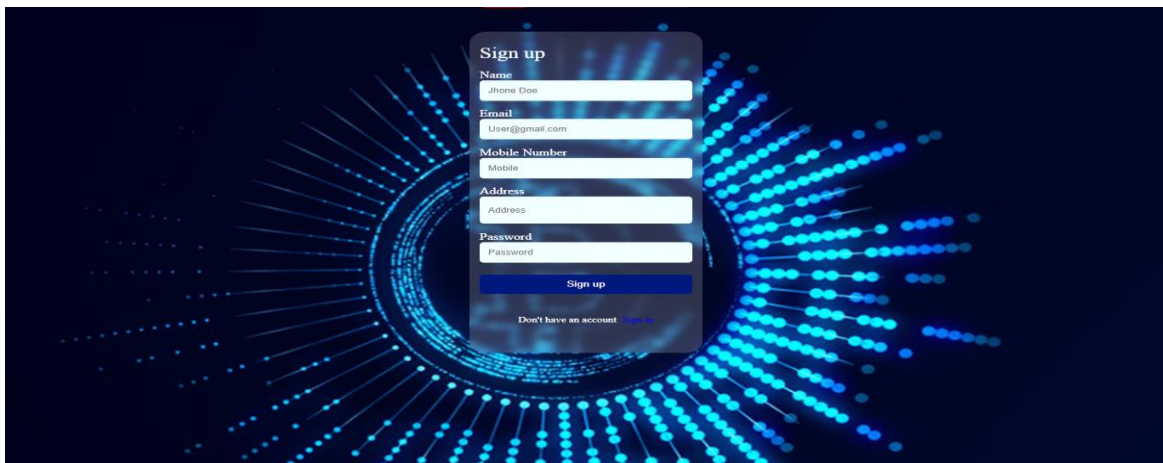
APPENDIX-B

SCREENSHOTS

Login page: User and admin login

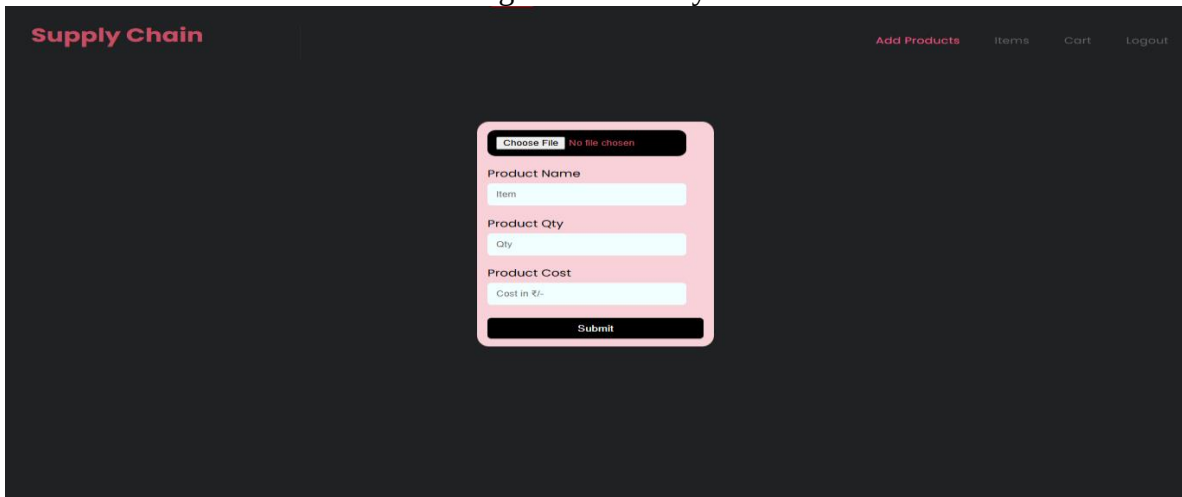


Sign Up: Here is the Page to Register



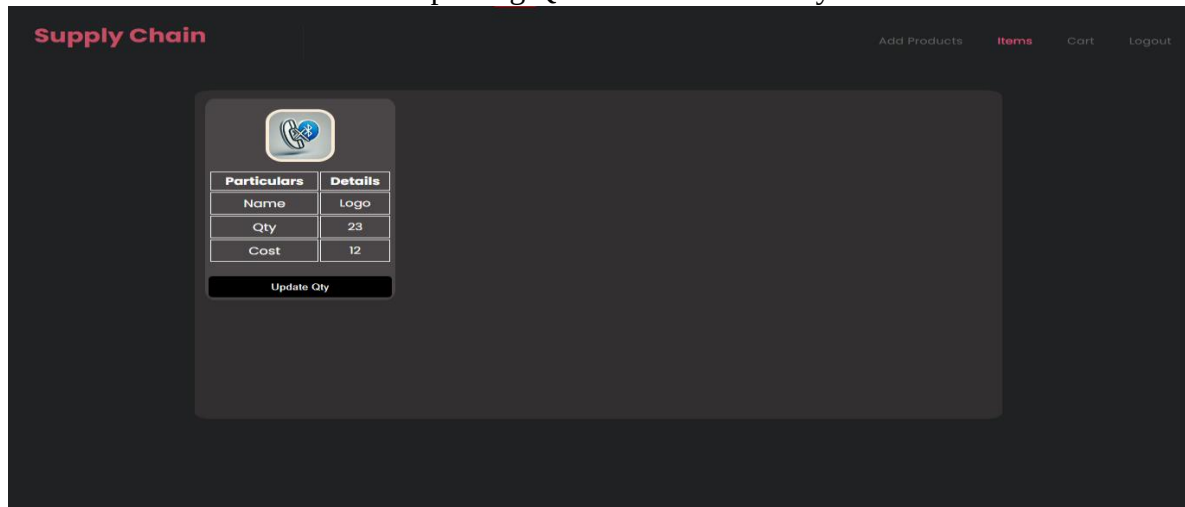
Admin Dashboard:

Here is the Admin dash board to manage for Inventory of the Products



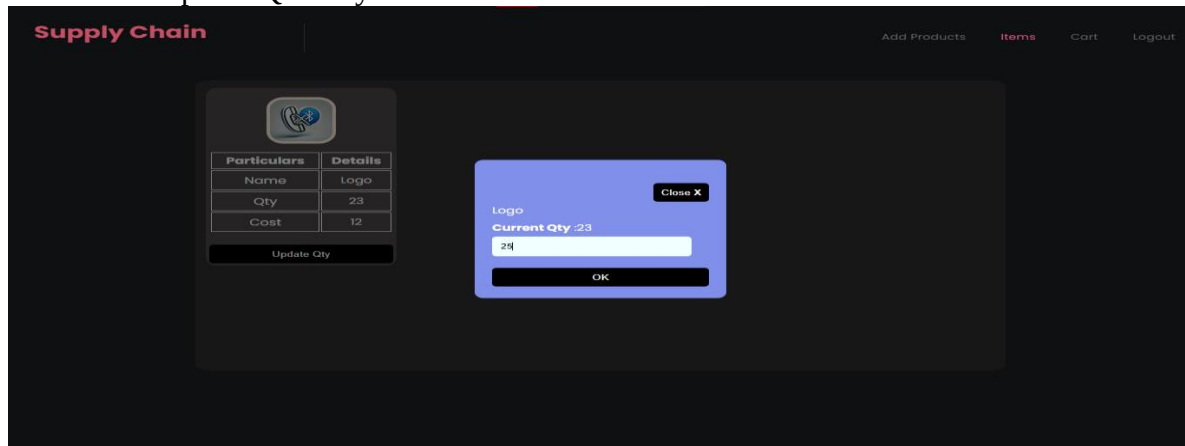
Admin View Items:

Admin Can view the Items for Updating Quantities of inventory



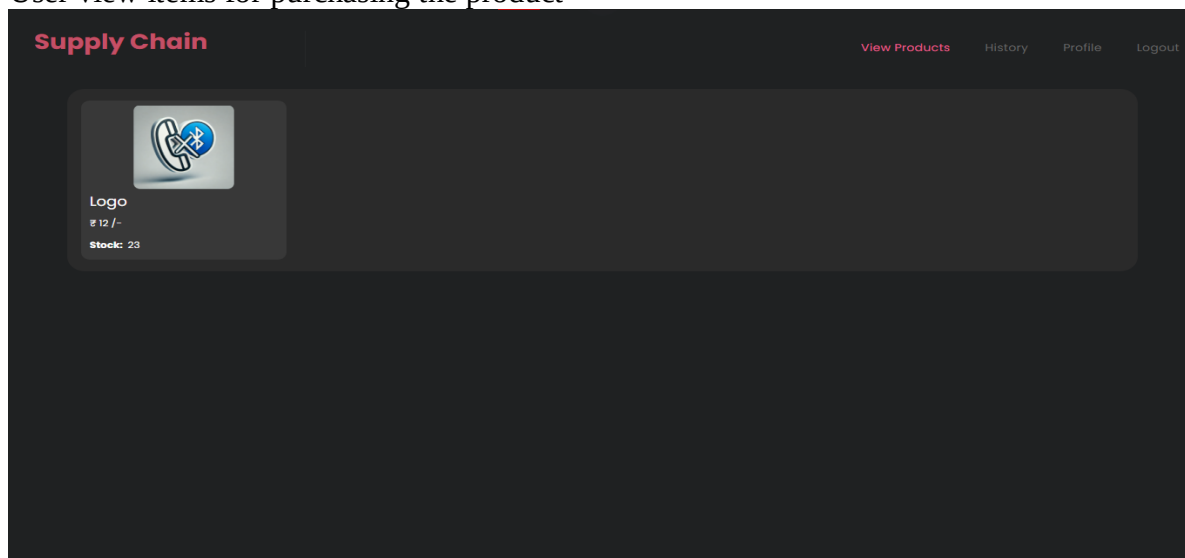
Admin update Qty:

Admin Can update Quantity of the Products



User Profile:

User view items for purchasing the product



Check out:

User checkout For Purchase the Product



The checkout form is displayed on a dark background. It features a 'Check out' title and a product image of a blue USB drive. Below the image, the form includes fields for 'Product Name' and 'Logo'. The 'Cost' is listed as ₹ 12/-, and the 'Quantity' is set to 4, with minus and plus buttons for adjustment. The 'Total' is ₹ 48/-. A 'Check Out' button is at the bottom.

Product Name	Logo
Cost	₹ 12/-
Quantity	4
Total	₹ 48/-

Check Out

Cart:

User can view the checkout for purchase



The 'Supply Chain' cart interface shows a 'Pending' status with a product image of a blue USB drive. The cart details are: Logo, Qty: 5, Cost: ₹ 12/-, and Total: ₹ 60/-.

Supply Chain

View Products History Profile Logout

Pending

Logo

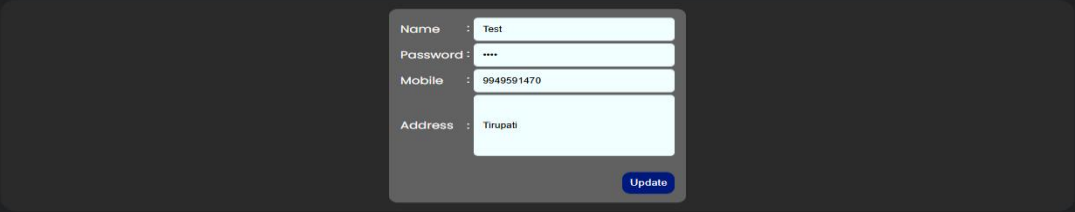
Qty: 5

Cost: ₹ 12/-

Total: ₹ 60/-

Profile Update:

Here user can update the profile



The 'Supply Chain' profile update interface shows a form with fields for Name, Password, Mobile, and Address. The current values are: Name: Test, Password: ****, Mobile: 9949591470, and Address: Tirupati. An 'Update' button is at the bottom right.

Supply Chain

View Products History Profile Logout

Name : Test

Password : ****

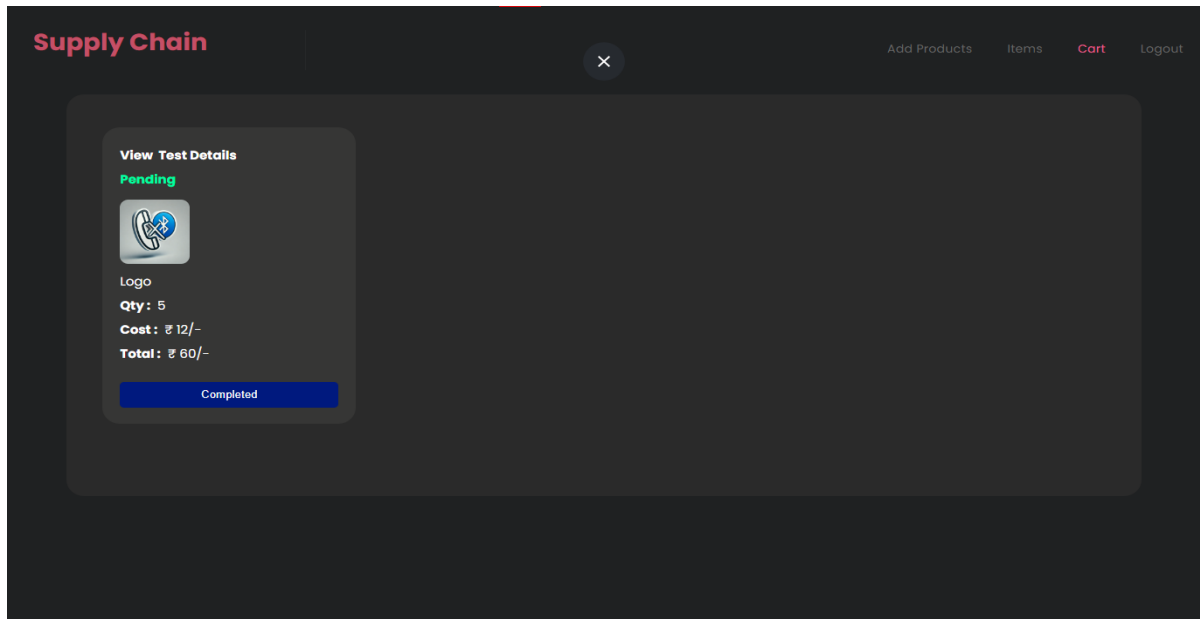
Mobile : 9949591470

Address : Tirupati

Update

Admin Cart Purchase:

Admin can view the purchase cart of the user



APPENDIX-C
ENCLOSURES
JOURNAL PUBLICATION CERTIFICATES

Journal of Xidian University <i>An UGC-CARE Approved Group - 2 Journal (Scopus Active Journal)</i> ISSN NO: 1001-2400 Scientific Journal Impact Factor - 5.4	
ACCEPTANCE LETTER TO AUTHOR	
Dear Author, With reference to your paper submitted "TRACK AND TRACE SUPPLYCHAIN", we are pleased to accept the same for publication in Journal of Xidian University. Manuscript ID: JXU-R11901 Please send the payment receipt for an online maintenance/processing fee of 2000+18%=2360 INR per paper. Please note that the amount we are charging is very nominal & only an online maintenance and processing fee. The Fee includes: • Online maintenance and processing charge. • Soft copy of certificate for each author. • No limitation of number of pages. • Editorial fee. Note: • Paper will be published within 48 hours after receiving the payment confirmation. • Once the paper is published online, corrections are not allowed. So send your final paper before publication. • Fee paid for publication of the paper will not be refunded under any circumstances. • In case of any query please do not hesitate to contact us at editorjxu@gmail.com. Early reply is appreciated. DATE 24-Dec-24	
Sincerely, Best regards, Jenny Corbett  http://www.xadzkjdx.cn/	

Trace and trace supply chain

ORIGINALITY REPORT

10%

SIMILARITY INDEX

5%

INTERNET SOURCES

8%

PUBLICATIONS

4%

STUDENT PAPERS

PRIMARY SOURCES

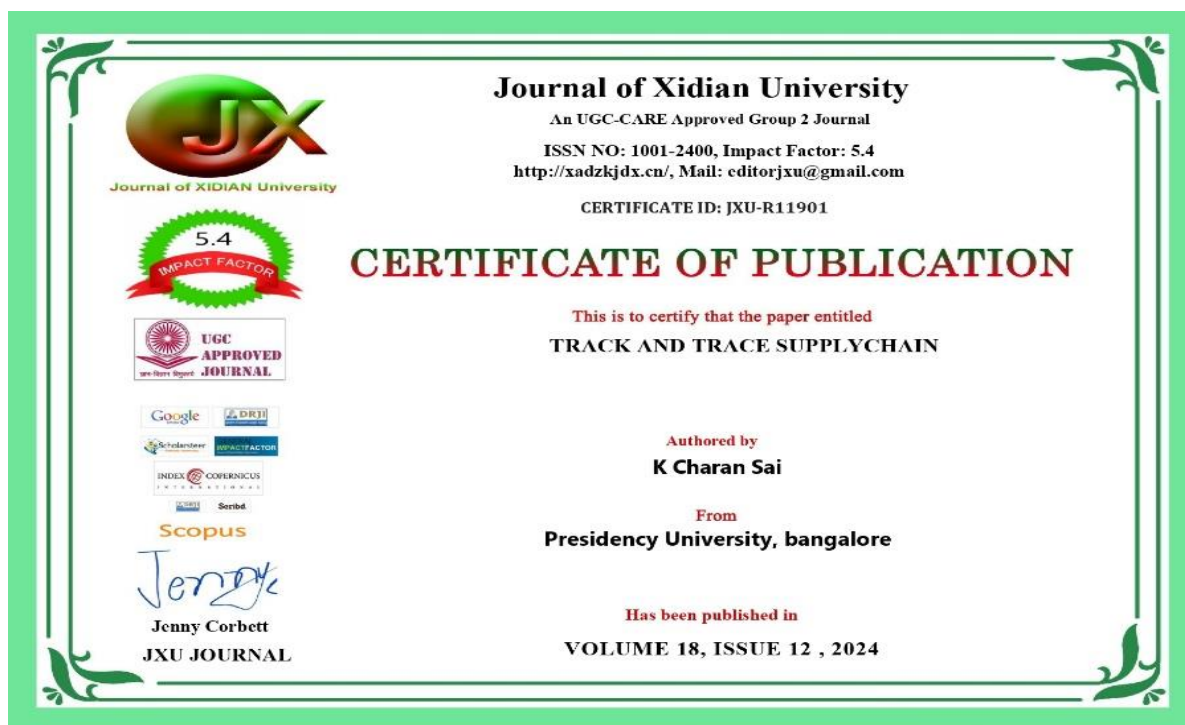
1	Santosh Shrivastava. "Recent trends in supply chain management of business-to-business firms: a review and future research directions", Journal of Business & Industrial Marketing, 2023 Publication	1%
2	ouci.dntb.gov.ua Internet Source	1%
3	Submitted to University of Greenwich Student Paper	1%
4	www.coursehero.com Internet Source	1%
5	www.researchgate.net Internet Source	1%
6	Submitted to The University of the South Pacific Student Paper	1%
7	blog.pacificcert.com Internet Source	1%











Sustainable Development Goals (SDGs)

